

Import Necessary Libraries

```
In [2]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Load the Dataset

```
In [3]: df=pd.read_csv("parkinsons.csv")
```

```
In [4]: df.head()
```

Out[4]:

	name	MDVP:Fo(Hz)	MDVP:Fhi(Hz)	MDVP:Flo(Hz)	MDVP:Jitter(%)	MDVP:Jitter(Abs)	MDVP:RAP	MDVP:PPQ	Jitter:DI
0	phon_R01_S01_1	119.992	157.302	74.997	0.00784	0.00007	0.00370	0.00554	0.011
1	phon_R01_S01_2	122.400	148.650	113.819	0.00968	0.00008	0.00465	0.00696	0.013
2	phon_R01_S01_3	116.682	131.111	111.555	0.01050	0.00009	0.00544	0.00781	0.016
3	phon_R01_S01_4	116.676	137.871	111.366	0.00997	0.00009	0.00502	0.00698	0.015
4	phon_R01_S01_5	116.014	141.781	110.655	0.01284	0.00011	0.00655	0.00908	0.019

5 rows × 24 columns

```
In [5]: df.shape
```

Out[5]: (195, 24)

Data Exploration and Cleaning

```
In [6]: df.describe()
```

Out[6]:

	MDVP:Fo(Hz)	MDVP:Fhi(Hz)	MDVP:Flo(Hz)	MDVP:Jitter(%)	MDVP:Jitter(Abs)	MDVP:RAP	MDVP:PPQ	Jitter:DDP	MDVP:Sh
count	195.000000	195.000000	195.000000	195.000000	195.000000	195.000000	195.000000	195.000000	195.000000
mean	154.228641	197.104918	116.324631	0.006220	0.000044	0.003306	0.003446	0.009920	0.000000
std	41.390065	91.491548	43.521413	0.004848	0.000035	0.002968	0.002759	0.008903	0.000000
min	88.333000	102.145000	65.476000	0.001680	0.000007	0.000680	0.000920	0.002040	0.000000
25%	117.572000	134.862500	84.291000	0.003460	0.000020	0.001660	0.001860	0.004985	0.000000
50%	148.790000	175.829000	104.315000	0.004940	0.000030	0.002500	0.002690	0.007490	0.000000
75%	182.769000	224.205500	140.018500	0.007365	0.000060	0.003835	0.003955	0.011505	0.000000
max	260.105000	592.030000	239.170000	0.033160	0.000260	0.021440	0.019580	0.064330	0.000000

8 rows × 23 columns

```
In [7]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 195 entries, 0 to 194
Data columns (total 24 columns):
#   Column                Non-Null Count  Dtype
---  -
0   name                  195 non-null   object
1   MDVP:Fo(Hz)          195 non-null   float64
2   MDVP:Fhi(Hz)         195 non-null   float64
3   MDVP:Flo(Hz)         195 non-null   float64
4   MDVP:Jitter(%)       195 non-null   float64
5   MDVP:Jitter(Abs)     195 non-null   float64
6   MDVP:RAP              195 non-null   float64
7   MDVP:PPQ             195 non-null   float64
8   Jitter:DDP           195 non-null   float64
9   MDVP:Shimmer          195 non-null   float64
10  MDVP:Shimmer(dB)      195 non-null   float64
11  Shimmer:APQ3          195 non-null   float64
12  Shimmer:APQ5          195 non-null   float64
13  MDVP:APQ              195 non-null   float64
14  Shimmer:DDA           195 non-null   float64
15  NHR                   195 non-null   float64
16  HNR                   195 non-null   float64
17  status                195 non-null   int64
18  RPDE                  195 non-null   float64
19  DFA                   195 non-null   float64
20  spread1               195 non-null   float64
21  spread2               195 non-null   float64
22  D2                    195 non-null   float64
23  PPE                   195 non-null   float64
dtypes: float64(22), int64(1), object(1)
memory usage: 36.7+ KB
```

```
In [8]: # Find null values in the data set
df.isnull().sum()
```

```
Out[8]: name                0
MDVP:Fo(Hz)                0
MDVP:Fhi(Hz)               0
MDVP:Flo(Hz)               0
MDVP:Jitter(%)             0
MDVP:Jitter(Abs)           0
MDVP:RAP                   0
MDVP:PPQ                   0
Jitter:DDP                 0
MDVP:Shimmer               0
MDVP:Shimmer(dB)           0
Shimmer:APQ3               0
Shimmer:APQ5               0
MDVP:APQ                   0
Shimmer:DDA                0
NHR                         0
HNR                         0
status                     0
RPDE                       0
DFA                        0
spread1                    0
spread2                    0
D2                         0
PPE                        0
dtype: int64
```

```
In [12]: # Fin finding unique values in the columns
for i in df.columns:
    print("*****",i,"*****")
    print(set(df[i].tolist()))
    print()
```

```
***** name *****
*****
{'phon_R01_S24_6', 'phon_R01_S05_1', 'phon_R01_S35_7', 'phon_R01_S07_6', 'phon_R01_S21_4', 'phon_R01_S35_3', 'ph
on_R01_S33_4', 'phon_R01_S08_6', 'phon_R01_S10_5', 'phon_R01_S02_6', 'phon_R01_S16_1', 'phon_R01_S34_4', 'phon_R
01_S01_2', 'phon_R01_S01_1', 'phon_R01_S16_3', 'phon_R01_S34_1', 'phon_R01_S34_2', 'phon_R01_S44_5', 'phon_R01_S
06_6', 'phon_R01_S10_1', 'phon_R01_S01_5', 'phon_R01_S20_6', 'phon_R01_S39_2', 'phon_R01_S42_4', 'phon_R01_S17_1
', 'phon_R01_S42_6', 'phon_R01_S26_4', 'phon_R01_S02_1', 'phon_R01_S21_6', 'phon_R01_S49_2', 'phon_R01_S06_4', '
phon_R01_S10_4', 'phon_R01_S13_5', 'phon_R01_S16_6', 'phon_R01_S33_6', 'phon_R01_S10_2', 'phon_R01_S20_5', 'phon
_R01_S27_4', 'phon_R01_S18_4', 'phon_R01_S43_4', 'phon_R01_S37_4', 'phon_R01_S22_6', 'phon_R01_S32_1', 'phon_R01
_S32_5', 'phon_R01_S39_3', 'phon_R01_S31_4', 'phon_R01_S43_2', 'phon_R01_S04_1', 'phon_R01_S07_2', 'phon_R01_S16
_5', 'phon_R01_S01_6', 'phon_R01_S31_2', 'phon_R01_S05_4', 'phon_R01_S07_4', 'phon_R01_S06_1', 'phon_R01_S04_3',
'phon_R01_S31_1', 'phon_R01_S44_3', 'phon_R01_S37_6', 'phon_R01_S05_2', 'phon_R01_S13_4', 'phon_R01_S19_4', 'pho
n_R01_S43_5', 'phon_R01_S31_6', 'phon_R01_S05_5', 'phon_R01_S04_5', 'phon_R01_S50_1', 'phon_R01_S33_3', 'phon_R0
1_S49_1', 'phon_R01_S27_6', 'phon_R01_S22_5', 'phon_R01_S39_4', 'phon_R01_S39_5', 'phon_R01_S26_5', 'phon_R01_S0
8_1', 'phon_R01_S33_1', 'phon_R01_S07_1', 'phon_R01_S13_6', 'phon_R01_S43_3', 'phon_R01_S35_2', 'phon_R01_S22_1'
, 'phon_R01_S44_4', 'phon_R01_S26_2', 'phon_R01_S31_5', 'phon_R01_S24_5', 'phon_R01_S50_4', 'phon_R01_S50_6', 'p
hon_R01_S02_3', 'phon_R01_S19_6', 'phon_R01_S02_2', 'phon_R01_S42_3', 'phon_R01_S24_1', 'phon_R01_S24_2', 'phon_
```

R01_S19_1', 'phon_R01_S33_2', 'phon_R01_S17_3', 'phon_R01_S20_4', 'phon_R01_S25_5', 'phon_R01_S25_2', 'phon_R01_S18_6', 'phon_R01_S27_7', 'phon_R01_S32_2', 'phon_R01_S37_1', 'phon_R01_S08_4', 'phon_R01_S22_2', 'phon_R01_S27_3', 'phon_R01_S32_3', 'phon_R01_S44_6', 'phon_R01_S18_3', 'phon_R01_S39_1', 'phon_R01_S18_1', 'phon_R01_S17_4', 'phon_R01_S01_3', 'phon_R01_S21_1', 'phon_R01_S17_5', 'phon_R01_S26_3', 'phon_R01_S49_3', 'phon_R01_S37_2', 'phon_R01_S26_6', 'phon_R01_S24_4', 'phon_R01_S34_3', 'phon_R01_S31_3', 'phon_R01_S19_2', 'phon_R01_S16_4', 'phon_R01_S08_3', 'phon_R01_S06_3', 'phon_R01_S27_5', 'phon_R01_S16_2', 'phon_R01_S05_6', 'phon_R01_S34_5', 'phon_R01_S33_5', 'phon_R01_S10_6', 'phon_R01_S22_4', 'phon_R01_S35_6', 'phon_R01_S21_3', 'phon_R01_S50_2', 'phon_R01_S02_5', 'phon_R01_S18_2', 'phon_R01_S27_1', 'phon_R01_S25_3', 'phon_R01_S21_7', 'phon_R01_S49_5', 'phon_R01_S39_6', 'phon_R01_S32_6', 'phon_R01_S49_6', 'phon_R01_S20_1', 'phon_R01_S20_2', 'phon_R01_S50_5', 'phon_R01_S25_6', 'phon_R01_S02_4', 'phon_R01_S13_2', 'phon_R01_S42_1', 'phon_R01_S19_3', 'phon_R01_S19_5', 'phon_R01_S07_3', 'phon_R01_S43_1', 'phon_R01_S44_1', 'phon_R01_S18_5', 'phon_R01_S50_3', 'phon_R01_S17_2', 'phon_R01_S49_4', 'phon_R01_S22_3', 'phon_R01_S24_3', 'phon_R01_S35_5', 'phon_R01_S25_1', 'phon_R01_S43_6', 'phon_R01_S27_2', 'phon_R01_S01_4', 'phon_R01_S06_2', 'phon_R01_S10_3', 'phon_R01_S05_3', 'phon_R01_S37_5', 'phon_R01_S20_3', 'phon_R01_S35_1', 'phon_R01_S07_5', 'phon_R01_S13_3', 'phon_R01_S06_5', 'phon_R01_S04_6', 'phon_R01_S21_2', 'phon_R01_S04_4', 'phon_R01_S04_2', 'phon_R01_S25_4', 'phon_R01_S26_1', 'phon_R01_S21_5', 'phon_R01_S08_2', 'phon_R01_S44_2', 'phon_R01_S32_4', 'phon_R01_S35_4', 'phon_R01_S42_2', 'phon_R01_S42_5', 'phon_R01_S34_6', 'phon_R01_S13_1', 'phon_R01_S08_5', 'phon_R01_S37_3', 'phon_R01_S17_6'}

***** MDVP:F0(Hz) *****

{102.273, 110.568, 110.453, 110.739, 112.239, 112.15, 112.547, 113.4, 113.166, 113.715, 114.238, 114.554, 114.563, 115.322, 115.38, 116.879, 116.15, 116.388, 116.848, 116.286, 117.274, 117.87, 117.963, 117.004, 117.226, 118.747, 119.031, 88.333, 119.056, 119.1, 91.904, 120.078, 120.289, 120.256, 95.056, 95.73, 95.385, 96.106, 95.605, 100.77, 100.96, 98.804, 121.345, 104.4, 122.336, 106.516, 107.332, 108.807, 109.86, 110.793, 110.707, 112.014, 112.876, 114.847, 110.417, 116.676, 116.014, 116.682, 119.992, 120.267, 120.08, 122.188, 122.964, 124.445, 120.552, 122.4, 126.344, 128.001, 129.336, 125.036, 125.791, 126.512, 125.641, 128.451, 128.94, 136.926, 136.969, 136.358, 139.173, 140.341, 139.224, 142.167, 143.533, 144.188, 142.729, 146.845, 138.19, 148.09, 148.272, 150.258, 151.955, 152.845, 153.046, 153.848, 153.88, 156.405, 155.358, 152.125, 157.821, 157.447, 159.116, 162.568, 163.656, 155.078, 158.219, 166.605, 167.93, 168.778, 166.888, 170.756, 171.041, 170.368, 173.917, 173.898, 169.774, 176.17, 177.876, 176.858, 178.222, 180.198, 180.978, 176.281, 179.711, 184.055, 178.285, 186.163, 187.733, 182.018, 138.145, 183.52, 188.62, 186.695, 193.03, 192.818, 197.076, 198.383, 199.228, 200.714, 201.464, 202.266, 203.184, 204.664, 198.458, 206.327, 202.805, 208.519, 209.144, 210.141, 208.083, 209.516, 214.289, 217.116, 145.174, 222.236, 223.365, 223.361, 228.832, 229.401, 228.969, 148.79, 148.143, 148.462, 236.2, 237.226, 149.689, 237.323, 240.301, 241.404, 242.852, 243.439, 244.99, 245.51, 150.44, 149.818, 151.884, 151.989, 151.872, 151.737, 252.455, 260.105, 154.003, 116.556, 156.239, 202.632, 174.188, 174.688, 176.824, 116.342, 126.144, 197.569, 198.116, 198.764, 201.774, 202.544, 127.93}

***** MDVP:F1(Hz) *****

{206.008, 131.669, 211.961, 565.74, 126.778, 217.627, 217.552, 581.289, 116.443, 586.567, 588.518, 592.03, 119.167, 128.143, 102.145, 102.305, 107.715, 108.664, 123.109, 110.019, 123.925, 112.24, 113.84, 112.777, 115.871, 115.697, 125.306, 125.213, 124.393, 120.103, 113.597, 122.611, 126.632, 123.723, 125.394, 126.358, 127.533, 128.611, 129.916, 130.049, 131.111, 131.162, 132.068, 134.231, 135.069, 134.656, 137.871, 137.244, 138.052, 139.867, 141.781, 139.71, 143.946, 140.557, 141.756, 141.068, 130.27, 148.65, 131.067, 131.897, 148.826, 150.449, 133.344, 154.609, 131.731, 128.442, 157.302, 157.765, 159.866, 159.774, 161.469, 162.215, 163.305, 162.824, 165.738, 166.607, 162.408, 163.335, 164.989, 163.267, 168.913, 172.86, 172.975, 135.738, 175.829, 176.595, 134.209, 177.291, 179.139, 129.038, 138.752, 142.369, 185.604, 142.83, 139.644, 189.398, 190.204, 191.759, 192.735, 193.221, 192.921, 195.107, 196.537, 197.724, 198.346, 197.173, 200.841, 201.249, 202.324, 200.125, 202.45, 205.56, 206.002, 206.896, 208.313, 208.701, 209.512, 211.604, 211.526, 210.565, 211.35, 215.203, 208.9, 217.455, 215.293, 219.29, 220.315, 221.3, 216.814, 223.982, 216.302, 225.93, 226.355, 227.383, 227.381, 224.429, 230.978, 231.345, 232.181, 232.706, 234.619, 233.481, 231.508, 237.494, 238.987, 239.541, 233.099, 241.35, 240.005, 243.709, 244.663, 245.135, 247.326, 248.834, 250.912, 252.221, 253.792, 253.017, 255.034, 126.609, 260.277, 261.487, 262.09, 263.872, 154.284, 264.919, 262.707, 268.796, 155.982, 271.314, 272.21, 157.339, 158.359, 127.349, 160.267, 160.368, 144.466, 161.078, 163.736, 163.441, 163.417, 133.374, 349.259, 396.961, 128.101, 127.611, 442.557, 442.824, 450.247, 479.697, 197.238, 198.109, 198.966, 492.892, 203.522}

***** MDVP:F1o(Hz) *****

{102.137, 104.437, 104.773, 104.095, 105.667, 105.554, 105.715, 106.821, 106.656, 107.816, 107.802, 108.634, 108.97, 109.216, 109.836, 109.815, 112.773, 112.173, 113.201, 65.75, 65.809, 65.782, 68.623, 67.021, 66.004, 67.343, 65.476, 68.401, 74.997, 75.836, 76.556, 77.63, 75.603, 78.128, 79.068, 76.779, 77.968, 82.764, 83.159, 84.072, 81.737, 86.292, 86.18, 80.055, 90.264, 91.226, 91.754, 85.545, 87.549, 95.628, 87.804, 96.206, 98.664, 99.77, 92.02, 93.978, 102.874, 103.37, 104.68, 105.007, 106.981, 107.024, 107.316, 108.153, 109.379, 110.402, 111.208, 104.315, 110.655, 111.366, 111.555, 113.819, 113.787, 114.82, 115.765, 114.676, 122.08, 117.495, 118.604, 125.61, 128.621, 129.859, 131.276, 132.857, 133.608, 133.751, 135.041, 138.99, 141.047, 142.822, 142.299, 144.878, 144.811, 144.148, 147.226, 148.691, 149.605, 149.442, 151.451, 144.736, 144.786, 155.495, 161.34, 163.564, 164.168, 165.982, 166.977, 168.013, 168.793, 173.015, 174.478, 175.456, 177.584, 177.258, 182.786, 185.258, 100.139, 189.621, 192.055, 192.091, 193.104, 195.708, 196.16, 197.079, 141.998, 199.02, 205.495, 219.783, 221.156, 223.634, 225.227, 227.911, 229.256, 231.848, 232.483, 232.435, 237.303, 239.17, 66.157, 100.757, 69.085, 71.948, 116.346, 74.677, 74.287, 74.904, 75.501, 75.344, 75.632, 75.349, 76.596, 77.022, 77.973, 78.032, 78.228, 79.032, 79.187, 79.82, 79.512, 79.543, 80.297, 80.637, 81.114, 82.063, 83.961, 83.34, 84.51, 85.902, 86.795, 86.232, 86.228, 86.647, 87.638, 88.251, 88.833, 89.686, 89.488, 90.794, 91.802, 91.121, 93.105, 100.673, 94.794, 94.246, 94.261, 95.654, 96.913, 96.983, 97.543, 97.527, 98.25, 99.923, 99.503, 116.187, 100.209}

***** MDVP:Jitter(%) *****

{0.00766, 0.00505, 0.00183, 0.00349, 0.00532, 0.0021, 0.00332, 0.00254, 0.00376, 0.00298, 0.00742, 0.00803, 0.00281, 0.00342, 0.01813, 0.00264, 0.00647, 0.00752, 0.00369, 0.00874, 0.00352, 0.00718, 0.0084, 0.01101, 0.00396, 0.01284, 0.00257, 0.00762, 0.0044, 0.00684, 0.00867, 0.00606, 0.00284, 0.0105, 0.00406, 0.00589, 0.00267, 0.00633, 0.00694, 0.00494, 0.00555, 0.00294, 0.00355, 0.02714, 0.00277, 0.0046, 0.00321, 0.00382, 0.00704, 0.03107, 0.00609, 0.00531, 0.00975, 0.0027, 0.00331, 0.00392, 0.0128, 0.00314, 0.00436, 0.00619, 0.00358, 0.01568, 0.0048, 0.00419, 0.01551, 0.00968, 0.00524, 0.00907, 0.00768, 0.00185, 0.0049, 0.00551, 0.00168, 0.0029, 0.01378, 0.00856, 0.00534, 0.00212, 0.00517, 0.01466, 0.00761, 0.005, 0.00178, 0.00544, 0.00605, 0.00788, 0.00405, 0.00727, 0.0

0.0071, 0.00205, 0.00266, 0.00327, 0.00571, 0.0031, 0.00432, 0.00293, 0.00476, 0.0052, 0.00459, 0.00198, 0.00581, 0.00842, 0.00381, 0.00442, 0.00564, 0.00303, 0.00747, 0.00225, 0.00608, 0.03011, 0.03316, 0.00757, 0.00496, 0.00174, 0.00235, 0.00296, 0.0074, 0.0054, 0.00923, 0.00784, 0.01872, 0.00462, 0.00567, 0.00428, 0.00289, 0.00733, 0.00411, 0.00533, 0.00472, 0.0136, 0.01038, 0.00333, 0.00455, 0.00516, 0.00638, 0.00316, 0.00238, 0.00621, 0.01936, 0.00282, 0.00404, 0.00709, 0.00448, 0.00831, 0.01719, 0.00248, 0.00692, 0.00309, 0.00431, 0.0037, 0.00492, 0.00997, 0.00353, 0.00397, 0.00336, 0.00841, 0.00519, 0.00258, 0.00702, 0.00502, 0.0018, 0.00241, 0.00346, 0.00407, 0.00651, 0.0039, 0.00451, 0.00817, 0.00495, 0.00356, 0.01627, 0.00417, 0.00339}

***** MDVP:Jitter(Abs) *****

{0.00011, 0.00022, 5e-05, 0.00016, 0.0001, 4e-05, 0.00015, 0.00026, 9e-06, 7e-06, 9e-05, 3e-05, 0.00014, 8e-05, 2e-05, 7e-05, 1e-05, 0.00012, 6e-05}

***** MDVP:RAP *****

{0.01854, 0.00244, 0.00366, 0.00105, 0.00166, 0.00349, 0.00593, 0.01159, 0.00393, 0.00254, 0.00115, 0.00176, 0.00237, 0.00159, 0.0022, 0.00281, 0.00403, 0.00647, 0.00186, 0.00247, 0.0043, 0.00169, 0.00996, 0.00291, 0.00352, 0.00152, 0.00135, 0.00196, 0.00257, 0.00118, 0.00284, 0.00467, 0.00406, 0.00206, 0.00389, 0.0045, 0.00189, 0.0025, 0.00372, 0.00233, 0.00294, 0.00094, 0.00155, 0.00277, 0.0026, 0.00321, 0.00826, 0.00121, 0.00182, 0.00165, 0.00226, 0.00287, 0.00209, 0.0027, 0.00331, 0.00131, 0.00114, 0.00175, 0.01568, 0.00863, 0.00219, 0.0028, 0.00463, 0.00141, 0.00202, 0.00263, 0.00124, 0.00507, 0.00368, 0.0049, 0.00168, 0.02144, 0.01117, 0.00351, 0.00412, 0.00534, 0.00334, 0.00134, 0.00117, 0.00178, 0.00622, 0.001, 0.00544, 0.00849, 0.00144, 0.00205, 0.00388, 0.00127, 0.00371, 0.00493, 0.00171, 0.00232, 0.00293, 0.00415, 0.00093, 0.00154, 0.00398, 0.00076, 0.00137, 0.00181, 0.00364, 0.00469, 0.00147, 0.00269, 0.00391, 0.00191, 0.00113, 0.00174, 0.00418, 0.00157, 0.00279, 0.00201, 0.00506, 0.00428, 0.00211, 0.00655, 0.00316, 0.00116, 0.00238, 0.00299, 0.00743, 0.0016, 0.00221, 0.00465, 0.00404, 0.00204, 0.00387, 0.01075, 0.0037, 0.00109, 0.0017, 0.00414, 0.00092, 0.00153, 0.00919, 0.00214, 0.00075, 0.00136, 0.0038, 0.00502, 0.0018, 0.00241, 0.00624, 0.00302, 0.00163, 0.00224, 0.018, 0.00146, 0.00268, 0.00068, 0.00373, 0.00173, 0.00295, 0.00356, 0.00905}

***** MDVP:PPQ *****

{0.00122, 0.00461, 0.00183, 0.00244, 0.00166, 0.00227, 0.00149, 0.00332, 0.00454, 0.00576, 0.00254, 0.00698, 0.00315, 0.00115, 0.00176, 0.00237, 0.0042, 0.00159, 0.0022, 0.00908, 0.00203, 0.00186, 0.00247, 0.00169, 0.00152, 0.00213, 0.00718, 0.00396, 0.00135, 0.00196, 0.00318, 0.0044, 0.00623, 0.00162, 0.00284, 0.00467, 0.00267, 0.00389, 0.0045, 0.00128, 0.00233, 0.00155, 0.00399, 0.00138, 0.00199, 0.00182, 0.01958, 0.00226, 0.00348, 0.00148, 0.0027, 0.00514, 0.00453, 0.00192, 0.00253, 0.00314, 0.00375, 0.00819, 0.00175, 0.00419, 0.0028, 0.00463, 0.00202, 0.00263, 0.00385, 0.00246, 0.00107, 0.00168, 0.0029, 0.00351, 0.00151, 0.00395, 0.00134, 0.00256, 0.00317, 0.003, 0.00422, 0.001, 0.00283, 0.00144, 0.00205, 0.00327, 0.01154, 0.00449, 0.00188, 0.00371, 0.00432, 0.00554, 0.00493, 0.00171, 0.00232, 0.00354, 0.00415, 0.00215, 0.00337, 0.00781, 0.00398, 0.00137, 0.0052, 0.00198, 0.00259, 0.00564, 0.00486, 0.002, 0.00469, 0.00147, 0.00208, 0.00269, 0.0033, 0.00696, 0.00113, 0.00235, 0.01628, 0.00096, 0.00218, 0.0034, 0.0014, 0.00184, 0.0075, 0.00428, 0.00106, 0.00167, 0.00289, 0.01699, 0.00211, 0.00655, 0.00133, 0.00194, 0.00316, 0.00238, 0.00221, 0.00909, 0.00387, 0.00448, 0.0017, 0.00231, 0.00292, 0.00092, 0.00153, 0.00275, 0.00336, 0.00136, 0.00197, 0.00258, 0.00963, 0.00319, 0.00946, 0.00241, 0.00485, 0.0099, 0.00346, 0.00207, 0.00329, 0.0039, 0.0019, 0.01522, 0.00312, 0.00173, 0.00234, 0.00539, 0.00478, 0.00339, 0.00139, 0.01027, 0.00261}

***** Jitter:DDP *****

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***** MDVP:Shimmer *****

{0.06511, 0.09419, 0.0605, 0.04701, 0.01681, 0.02047, 0.02752, 0.01098, 0.02308, 0.05233, 0.01725, 0.04128, 0.0203, 0.01064, 0.02857, 0.01752, 0.01613, 0.05643, 0.01169, 0.0103, 0.02362, 0.01152, 0.01718, 0.01457, 0.03111, 0.03999, 0.02223, 0.09178, 0.01657, 0.02145, 0.01884, 0.01745, 0.04087, 0.01484, 0.04192, 0.02033, 0.03121, 0.06734, 0.0145, 0.04009, 0.02199, 0.03026, 0.01494, 0.06134, 0.03209, 0.03087, 0.01033, 0.01843, 0.0166, 0.05517, 0.01643, 0.06727, 0.01043, 0.04351, 0.04978, 0.03202, 0.04795, 0.02297, 0.03995, 0.01131, 0.01897, 0.0168, 0.01419, 0.01358, 0.01663, 0.03273, 0.02751, 0.00958, 0.01463, 0.02551, 0.01263, 0.06425, 0.02534, 0.01185, 0.02378, 0.03327, 0.04137, 0.01412, 0.03527, 0.01795, 0.01761, 0.013, 0.02293, 0.03198, 0.03381, 0.03886, 0.01022, 0.05279, 0.02215, 0.02093, 0.04374, 0.03225, 0.05384, 0.03852, 0.04313, 0.04879, 0.08143, 0.02442, 0.02503, 0.01659, 0.01015, 0.00954, 0.05428, 0.02286, 0.04479, 0.03235, 0.01259, 0.01642, 0.02791, 0.02852, 0.01503, 0.04689, 0.02008, 0.0717, 0.01242, 0.01564, 0.02574, 0.01608, 0.01791, 0.01469, 0.03767, 0.0617, 0.02296, 0.02662, 0.02018, 0.01574, 0.05492, 0.02184, 0.02645, 0.03716, 0.03272, 0.01279, 0.01906, 0.01706, 0.01201, 0.02838, 0.01828, 0.01567, 0.11908, 0.02682, 0.02177, 0.01472, 0.01194, 0.02343, 0.01516, 0.02143, 0.02448, 0.0176, 0.01299, 0.02126, 0.04024, 0.07959, 0.04912, 0.03658, 0.02814, 0.01909, 0.02719, 0.03485, 0.0419, 0.02536, 0.01831, 0.01919, 0.01997, 0.01458, 0.03156, 0.01851, 0.05925, 0.01346, 0.04932, 0.01024, 0.06725, 0.07118, 0.08684, 0.03044, 0.0381, 0.0583, 0.01756, 0.01495, 0.03715, 0.01861, 0.02427, 0.03759, 0.0664, 0.02105, 0.01966, 0.01644, 0.01444, 0.02498}

***** MDVP:Shimmer(dB) *****

{0.482, 0.542, 0.135, 0.478, 0.129, 0.377, 0.361, 0.383, 0.381, 0.131, 0.125, 0.369, 0.111, 0.107, 0.103, 0.117, 0.283, 1.302, 0.267, 0.136, 0.431, 0.181, 0.21, 0.226, 0.097, 0.089, 0.228, 0.206, 0.216, 0.093, 0.224, 0.099, 0.476, 0.175, 0.659, 0.784, 0.212, 0.249, 0.124, 0.085, 0.618, 0.772, 0.833, 0.272, 0.442, 0.237, 0.426, 0.456, 0.186, 0.202, 0.327, 0.331, 0.313, 0.198, 0.307, 0.192, 0.891, 0.223, 0.571, 0.821, 0.18, 0.217, 0.379, 0.154, 0.158, 0.164, 0.168, 0.17, 0.152, 0.148, 0.142, 0.441, 0.236, 0.265, 0.339, 0.634, 0.171, 0.626, 0.584, 0.517, 0.14, 0.134, 0.255, 0.497, 0.263, 0.132, 0.35, 0.276, 0.93, 0.241, 0.112, 0.106, 0.116, 0.19, 0.235, 0.221, 0.094, 0.207, 0.209, 0.231, 0.233, 0.098, 0.09, 0.225, 0.348, 0.133, 0.422, 0.297, 0.281, 0.406, 0.246, 0.65, 0.275, 0.365, 0.58, 0.191, 0.308, 0.334, 0.328, 0.296, 0.197, 0.185, 0.435, 0.189, 0.342, 0.138, 0.257, 0.722, 0.126, 0.405, 0.325, 0.37, 0.143, 0.161, 0.151, 0.141, 0.145, 0.165, 0.155, 0.149, 0.163, 0.364, 0.438, 1.018, 0.483, 0.266, 0.637, 0.137, 0.256}

***** Shimmer:APQ3 *****

{0.02542, 0.0081, 0.01454, 0.00793, 0.03152, 0.03474, 0.01176, 0.01803, 0.01864, 0.02413, 0.04284, 0.00942, 0.00881, 0.0082, 0.01064, 0.0183, 0.00742, 0.01186, 0.00864, 0.0365, 0.01813, 0.02135, 0.03223, 0.00847, 0.00664, 0.00586, 0.00725, 0.00969, 0.00769, 0.01396, 0.01013, 0.00952, 0.04016, 0.01579, 0.00874, 0.02328, 0.01379, 0.00796, 0.01806, 0.00779, 0.01484, 0.01667, 0.01789, 0.0064, 0.01284, 0.01084, 0.01006, 0.05358, 0.01189, 0.00867, 0.01372, 0.00606, 0.00667, 0.00728, 0.02182, 0.05551, 0.01155, 0.00772, 0.01721, 0.01277, 0.00633, 0.0307, 0.00938, 0.01321, 0.00738, 0.00538, 0.0066, 0.00721, 0.01792, 0.03341, 0.01026, 0.02924, 0.02297, 0.00504, 0.00748, 0.0349, 0.01192, 0.01514, 0.02385, 0.00975, 0.03134, 0.02107, 0.02229, 0.02073, 0.03788, 0.01107, 0.02683, 0.00829, 0.01073, 0.0049, 0.00812, 0.02266, 0.01117, 0.015, 0.00534, 0.00656, 0.01483, 0.01805, 0.00839, 0.00883, 0.01205, 0.02032, 0.01771, 0.01432, 0.01371, 0.00849, 0.01154, 0.01659, 0.00754, 0.03357, 0.00476, 0.02896, 0.01547, 0.02757, 0.00703, 0.02679, 0.01713, 0.0093, 0.00469, 0.04421, 0.02062, 0.03611, 0.01035, 0.00974, 0.01235, 0.00774, 0.01279, 0.01079, 0.0134, 0.00696, 0.02228, 0.00757, 0.03804, 0.01323, 0.00557, 0.00679, 0.02055, 0.02699, 0.00906, 0.01289, 0.00967, 0.02865, 0.00889, 0.01394, 0.0095, 0.02021, 0.01638, 0.02587, 0.00855, 0.0136, 0.02187, 0.01604, 0.00777, 0.00455, 0.01143, 0.00882, 0.01265, 0.02336, 0.00726, 0.01475, 0.00631, 0.01441, 0.00614, 0.01241, 0.01424, 0.01868, 0.00563, 0.01773, 0.01268, 0.01373, 0.03671, 0.05647, 0.00468, 0.02383, 0.02749, 0.03515, 0.02471, 0.0261, 0.01644, 0.00617, 0.00861, 0.01732, 0.00522, 0.0141}

***** Shimmer:APQ5 *****

{0.04518, 0.04962, 0.00888, 0.02498, 0.00932, 0.0061, 0.01759, 0.04101, 0.0263, 0.0142, 0.00776, 0.01342, 0.00898, 0.00959, 0.00576, 0.01725, 0.01325, 0.0082, 0.01003, 0.00925, 0.01108, 0.01169, 0.0123, 0.00786, 0.03572, 0.0063, 0.05426, 0.01657, 0.01057, 0.01223, 0.00901, 0.02494, 0.01284, 0.00701, 0.0165, 0.02521, 0.00606, 0.00789, 0.00972, 0.02321, 0.03714, 0.01033, 0.01277, 0.01399, 0.0247, 0.01199, 0.02592, 0.01582, 0.01121, 0.01321, 0.01426, 0.01365, 0.00721, 0.01609, 0.01992, 0.00582, 0.00948, 0.05005, 0.01375, 0.02768, 0.01558, 0.01219, 0.01341, 0.00941, 0.0068, 0.04005, 0.00802, 0.0794, 0.03022, 0.01812, 0.02422, 0.01012, 0.02161, 0.04825, 0.02466, 0.01439, 0.01117, 0.01483, 0.01805, 0.00717, 0.01161, 0.02493, 0.04791, 0.01405, 0.01144, 0.01893, 0.02415, 0.00744, 0.01815, 0.03347, 0.0353, 0.02076, 0.00788, 0.00971, 0.01859, 0.00588, 0.00832, 0.0313, 0.00632, 0.01459, 0.00815, 0.00937, 0.02591, 0.01886, 0.01964, 0.01581, 0.00876, 0.01625, 0.00659, 0.01103, 0.02374, 0.01347, 0.02174, 0.0254, 0.00825, 0.01974, 0.00886, 0.00747, 0.01191, 0.01574, 0.0073, 0.01296, 0.04282, 0.03794, 0.03672, 0.02567, 0.0194, 0.02245, 0.00957, 0.00879, 0.01906, 0.00818, 0.04265, 0.01062, 0.03526, 0.01994, 0.01289, 0.01272, 0.0095, 0.01072, 0.0458, 0.00933, 0.01177, 0.00977, 0.03858, 0.0116, 0.01038, 0.01421, 0.01343, 0.03963, 0.05556, 0.01804, 0.02231, 0.01909, 0.0076, 0.01021, 0.00621, 0.01553, 0.02302, 0.01075, 0.01841, 0.00631, 0.0057, 0.01258, 0.03112, 0.01641, 0.02024, 0.0158, 0.02973, 0.01058, 0.02451, 0.00641, 0.01024, 0.00946, 0.00885, 0.04254, 0.01495, 0.0099, 0.02383, 0.01478, 0.01783, 0.04159, 0.01722, 0.01017, 0.00956, 0.012, 0.04003, 0.0181, 0.00905}

***** MDVP:APQ *****

{0.0343, 0.01149, 0.01715, 0.13778, 0.01271, 0.02603, 0.00871, 0.00993, 0.03091, 0.01559, 0.01359, 0.00915, 0.0046, 0.03772, 0.02074, 0.01491, 0.01691, 0.01552, 0.0123, 0.02301, 0.02745, 0.02084, 0.01318, 0.01179, 0.02067, 0.05114, 0.01301, 0.02006, 0.01345, 0.02877, 0.01667, 0.00762, 0.03243, 0.01267, 0.00928, 0.01433, 0.01799, 0.01233, 0.01677, 0.03392, 0.01033, 0.02704, 0.04802, 0.0569, 0.02809, 0.04114, 0.02931, 0.01016, 0.04368, 0.04246, 0.01826, 0.02148, 0.01382, 0.0086, 0.01948, 0.01104, 0.01931, 0.02802, 0.01148, 0.01009, 0.01331, 0.01636, 0.04683, 0.04134, 0.08808, 0.02402, 0.01758, 0.01497, 0.01263, 0.0431, 0.00802, 0.06259, 0.02073, 0.01246, 0.01307, 0.01612, 0.01751, 0.02056, 0.01351, 0.0351, 0.04398, 0.01151, 0.01734, 0.00951, 0.01717, 0.01256, 0.02971, 0.01944, 0.01439, 0.02571, 0.02876, 0.05174, 0.01344, 0.01666, 0.01144, 0.01588, 0.01771, 0.05767, 0.02137, 0.02259, 0.0131, 0.02764, 0.03635, 0.0172, 0.0253, 0.01059, 0.01947, 0.01852, 0.00903, 0.01652, 0.02157, 0.02784, 0.02018, 0.04055, 0.03455, 0.06824, 0.01879, 0.02428, 0.00957, 0.0134, 0.04465, 0.0114, 0.03316, 0.02916, 0.02455, 0.01767, 0.01506, 0.01367, 0.01872, 0.0437, 0.01133, 0.03736, 0.00811, 0.01194, 0.01255, 0.0277, 0.01621, 0.0378, 0.01604, 0.01909, 0.00882, 0.01892, 0.02214, 0.01831, 0.05783, 0.04451, 0.01309, 0.02519, 0.00726, 0.0359, 0.01797, 0.01614, 0.03651, 0.02824, 0.06359, 0.01014, 0.03051, 0.01075, 0.01685, 0.0219, 0.01363, 0.00719, 0.08318, 0.01285, 0.0119, 0.01251, 0.01956, 0.02139, 0.01312, 0.03105, 0.01756, 0.01373, 0.02444, 0.01234, 0.02949, 0.03088, 0.01095, 0.04525, 0.01661, 0.06196, 0.04464, 0.014, 0.012, 0.02454, 0.03908, 0.06023, 0.0322, 0.01366, 0.01949}

***** Shimmer:DDA *****

{0.0827, 0.03969, 0.06799, 0.11012, 0.06406, 0.01471, 0.02542, 0.02925, 0.01898, 0.02908, 0.02647, 0.02308, 0.01603, 0.01403, 0.0445, 0.06321, 0.10422, 0.03867, 0.06165, 0.02518, 0.01979, 0.02623, 0.06985, 0.13262, 0.08595, 0.05592, 0.0265, 0.02789, 0.06097, 0.06219, 0.02389, 0.04426, 0.01406, 0.04914, 0.02921, 0.08771, 0.06185, 0.07761, 0.03104, 0.02643, 0.0246, 0.02321, 0.05368, 0.0549, 0.04114, 0.04019, 0.03836, 0.02226, 0.02487, 0.02548, 0.04324, 0.09669, 0.03253, 0.02592, 0.0227, 0.03463, 0.03724, 0.05439, 0.01992, 0.02436, 0.02175, 0.04812, 0.01758, 0.03429, 0.03568, 0.02602, 0.05605, 0.07625, 0.07154, 0.11411, 0.02307, 0.02429, 0.01968, 0.04188, 0.03039, 0.09455, 0.0532, 0.16654, 0.08689, 0.09211, 0.08096, 0.16942, 0.07008, 0.05164, 0.04137, 0.02666, 0.0412, 0.05408, 0.11363, 0.0715, 0.03615, 0.02849, 0.06062, 0.0332, 0.0211, 0.03964, 0.02337, 0.07238, 0.03852, 0.10024, 0.16074, 0.10546, 0.12851, 0.03191, 0.02164, 0.04079, 0.03557, 0.03218, 0.0233, 0.05377, 0.06892, 0.01364, 0.04933, 0.03529, 0.04272, 0.03706, 0.04977, 0.01513, 0.02184, 0.10949, 0.01818, 0.04543, 0.12047, 0.0335, 0.04499, 0.02001, 0.02245, 0.03794, 0.05414, 0.02855, 0.02089, 0.03804, 0.01689, 0.01567, 0.01428, 0.02316, 0.06685, 0.05197, 0.03831, 0.04736, 0.04231, 0.01672, 0.02177, 0.02038, 0.04641, 0.09403, 0.02814, 0.08037, 0.05312, 0.06688, 0.04363, 0.02214, 0.02719, 0.05417, 0.04451, 0.05139, 0.01892, 0.02902, 0.05, 0.0238, 0.01919, 0.04295, 0.08247, 0.03851, 0.10833, 0.01614, 0.01841, 0.0783, 0.04322, 0.04183, 0.03017, 0.03078, 0.07413, 0.01407, 0.01851, 0.02583, 0.04115, 0.02566, 0.02827, 0.02261, 0.06545, 0.1047, 0.06562, 0.02488, 0.1007, 0.03576, 0.0805, 0.03237, 0.03464}

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***** NHR *****
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, 0.00786, 0.00586, 0.00969, 0.01796, 0.0083, 0.0184, 0.0547, 0.08151, 0.16265, 0.0043, 0.01823, 0.01179, 0.0047
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***** PPE *****

{0.377429, 0.316395, 0.3167, 0.259765, 0.095032, 0.105993, 0.108514, 0.085569, 0.093534, 0.365391, 0.301952, 0.27727, 0.214346, 0.164916, 0.183572, 0.056141, 0.270173, 0.220546, 0.216638, 0.170633, 0.103224, 0.113942, 0.209863, 0.112878, 0.115515, 0.378483, 0.220657, 0.19273, 0.086398, 0.119308, 0.231571, 0.138512, 0.199889, 0.141929, 0.242119, 0.160809, 0.229892, 0.24308, 0.244948, 0.169923, 0.215724, 0.121777, 0.231723, 0.09632, 0.102706, 0.11513, 0.13639, 0.277948, 0.251972, 0.093193, 0.160376, 0.174152, 0.214075, 0.165827, 0.274407, 0.284654, 0.368674, 0.357775, 0.275931, 0.228624, 0.234589, 0.430788, 0.09147, 0.098555, 0.147403, 0.180828, 0.327978, 0.160812, 0.05761, 0.09622, 0.185668, 0.209191, 0.247455, 0.212386, 0.268144, 0.370961, 0.454721, 0.1701, 0.068501, 0.160691, 0.225461, 0.186489, 0.215558, 0.222716, 0.196535, 0.228319, 0.340623, 0.179677, 0.28278, 0.418646, 0.131728, 0.367233, 0.144105, 0.073581, 0.091546, 0.105306, 0.141422, 0.132703, 0.219514, 0.249703, 0.261305, 0.189032, 0.168895, 0.332634, 0.410335, 0.271362, 0.253556, 0.128872, 0.162999, 0.11573, 0.344834, 0.242981, 0.260015, 0.112838, 0.163118, 0.301487, 0.044539, 0.260375, 0.206256, 0.238281, 0.457533, 0.170106, 0.177807, 0.119652, 0.184985, 0.194052, 0.23252, 0.226247, 0.173218, 0.097336, 0.163755, 0.24974, 0.215961, 0.135242, 0.18558, 0.177275, 0.151709, 0.274387, 0.19771, 0.444774, 0.232861, 0.336085, 0.168581, 0.22052, 0.156368, 0.106802, 0.112856, 0.07587, 0.120605, 0.138868, 0.335041, 0.260633, 0.259451, 0.101516, 0.368975, 0.211756, 0.322111, 0.314464, 0.285695, 0.326197, 0.244512, 0.130554, 0.232744, 0.264666, 0.527367, 0.174429, 0.184067, 0.181988, 0.356881, 0.100881, 0.232209, 0.220968, 0.117399, 0.141958, 0.147491, 0.193918, 0.218164, 0.226156, 0.18818, 0.200423, 0.144614, 0.13412, 0.13305, 0.103561, 0.152428, 0.091604, 0.133867, 0.123306, 0.332086, 0.234809, 0.160306, 0.202879, 0.252404, 0.148569, 0.159777}

```
In [13]: df["status"].value_counts()
```

```
Out[13]: status
1      147
0       48
Name: count, dtype: int64
```

EDA

```
In [22]: # Find the distribution of the data
def distplot(col):
    sns.distplot(df[col])
    plt.show()
for i in list(df.columns)[1:]:
    distplot(i)
```

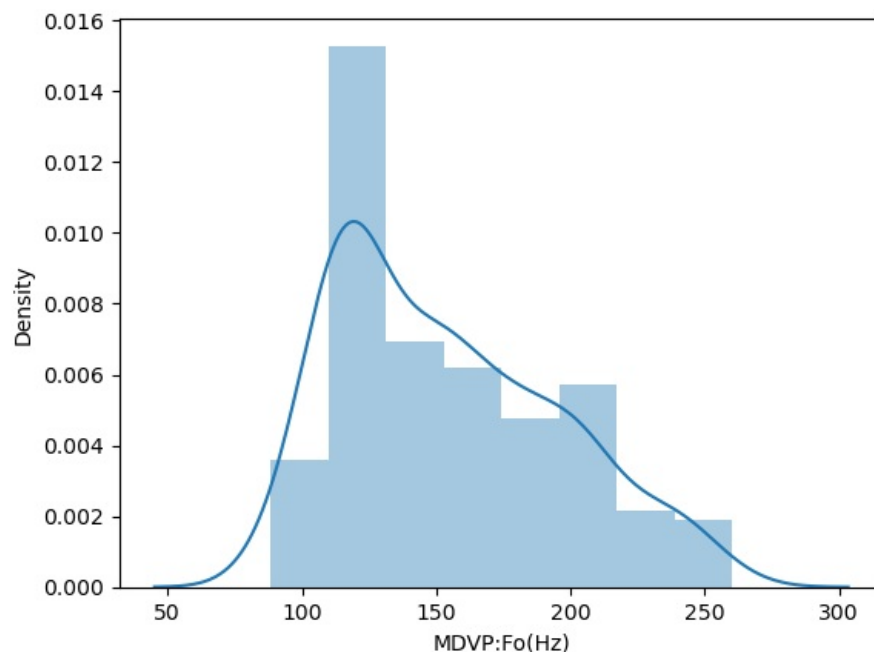
C:\Users\abhis\AppData\Local\Temp\ipykernel_14280\3978003669.py:3: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see <https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751>

```
sns.distplot(df[col])
```



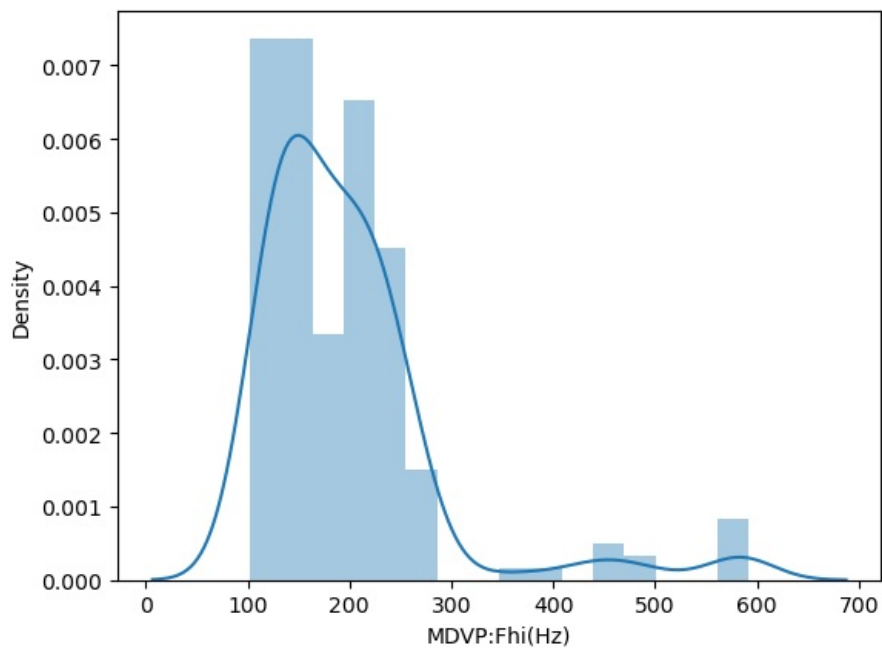
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```
sns.distplot(df[col])
```



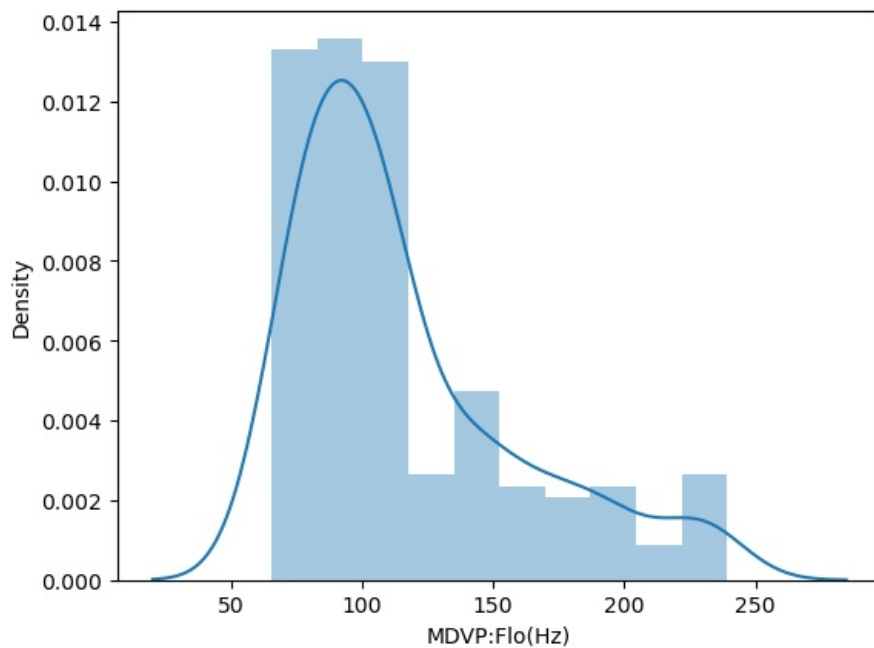
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```
sns.distplot(df[col])
```



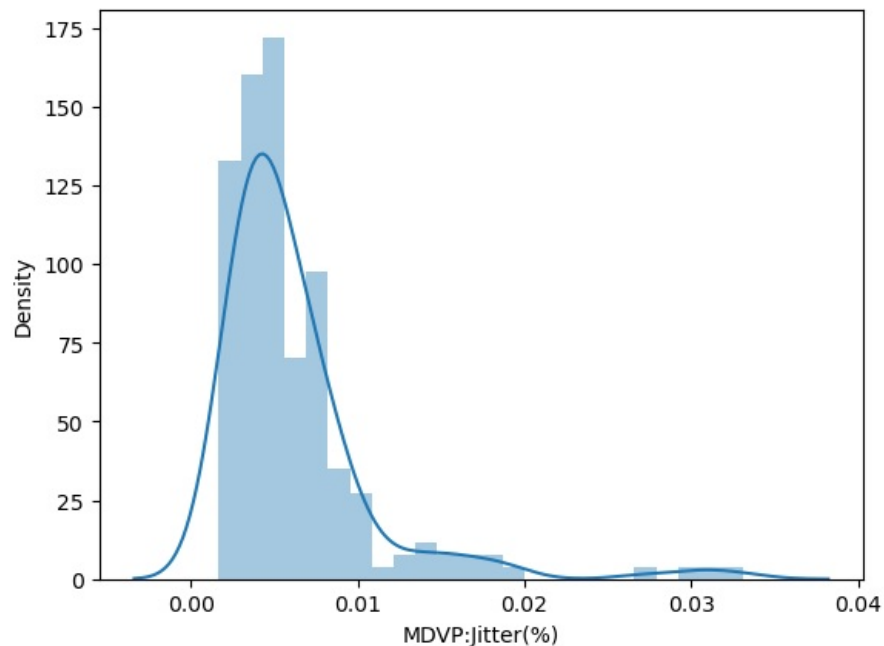
C:\Users\abhis\AppData\Local\Temp\ipykernel_14280\3978003669.py:3: UserWarning:

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```
sns.distplot(df[col])
```



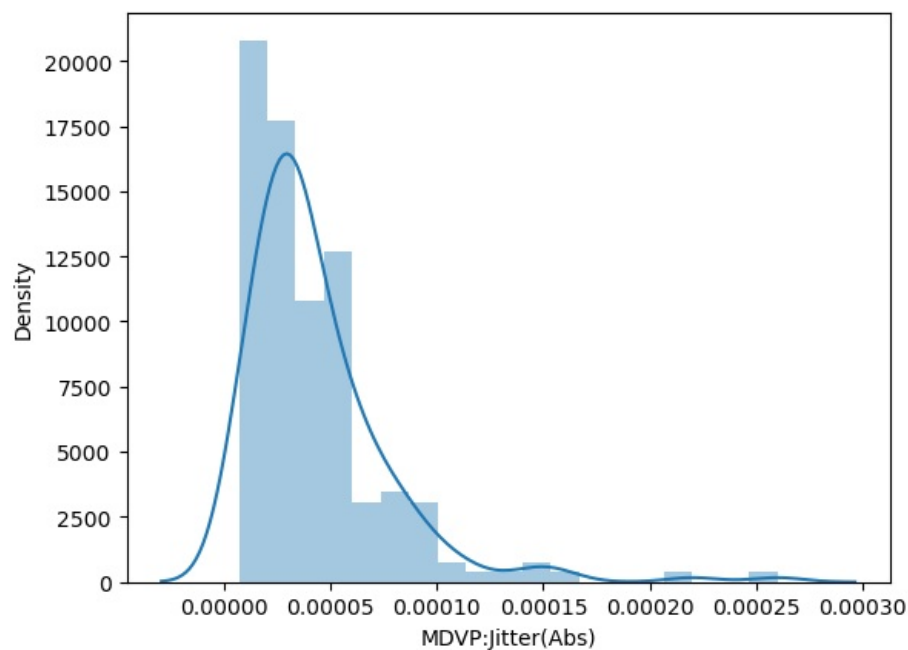
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```



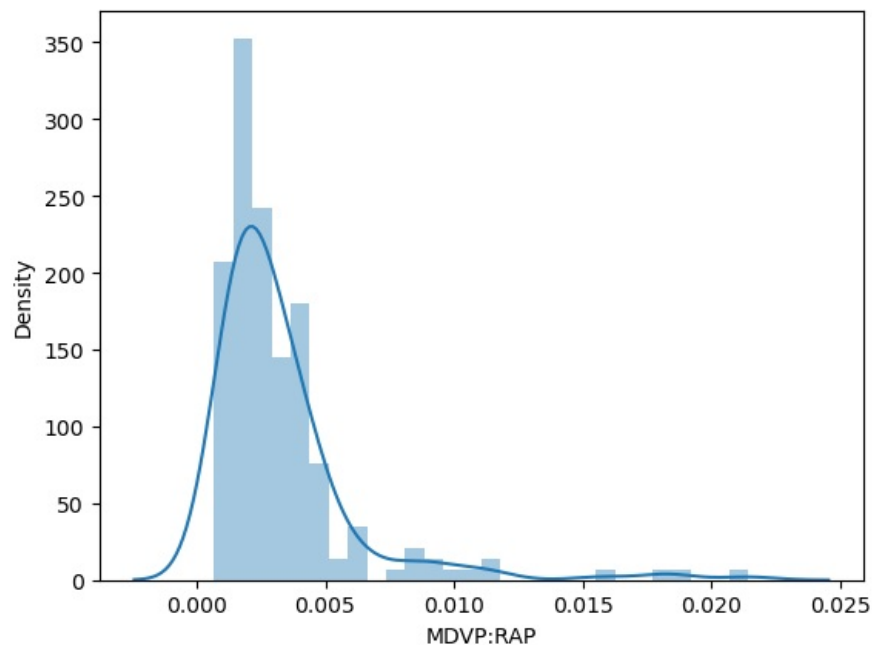
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```
sns.distplot(df[col])
```



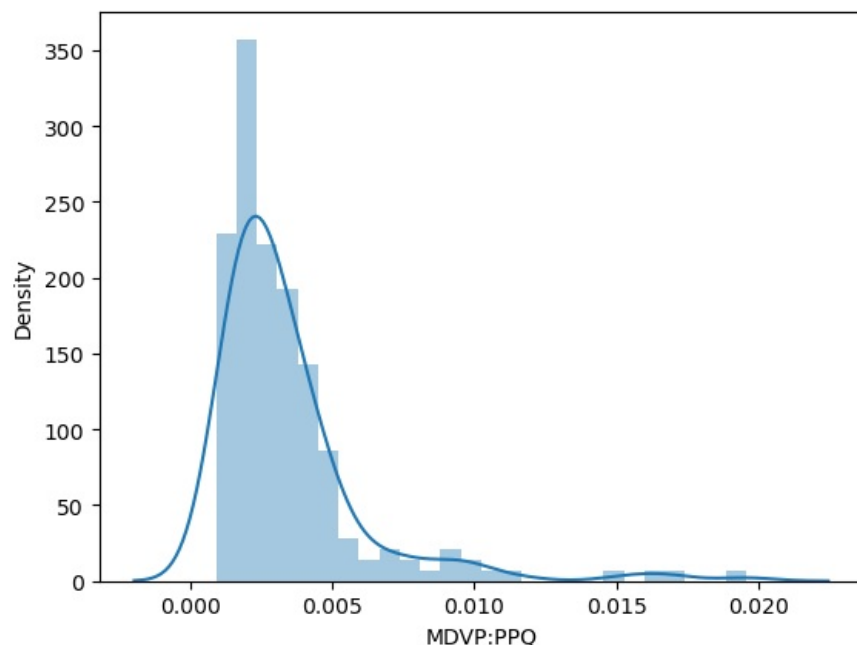
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```



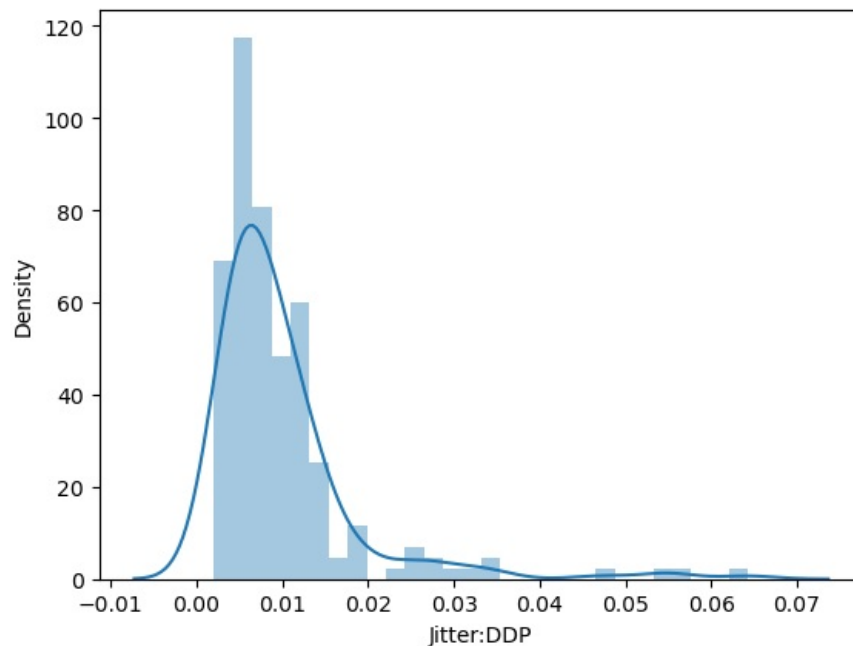
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```



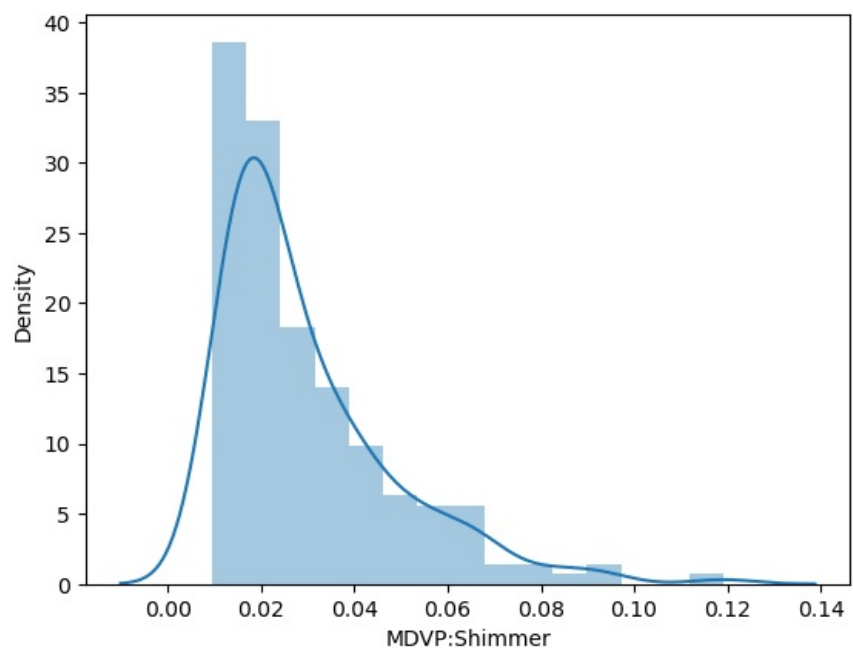
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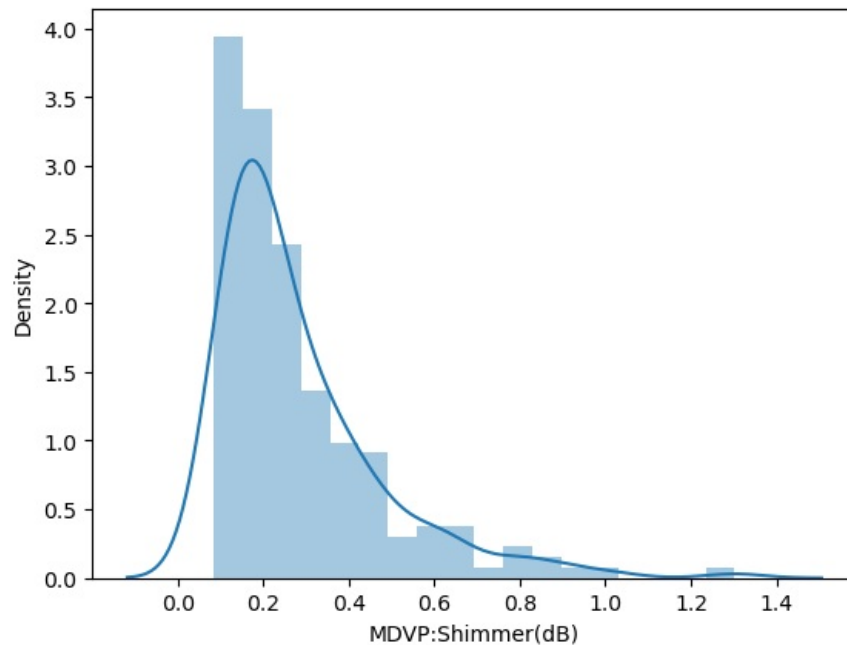
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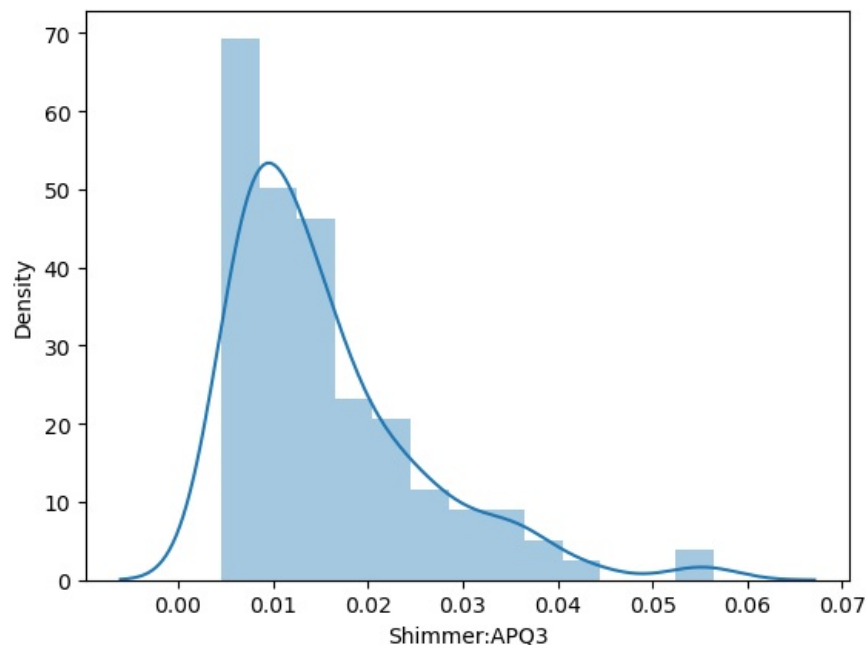
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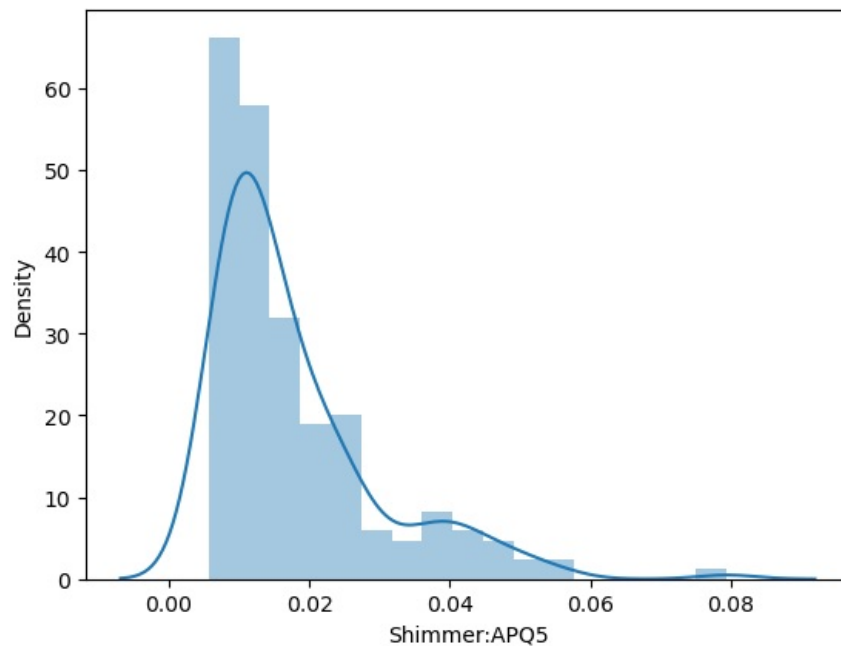
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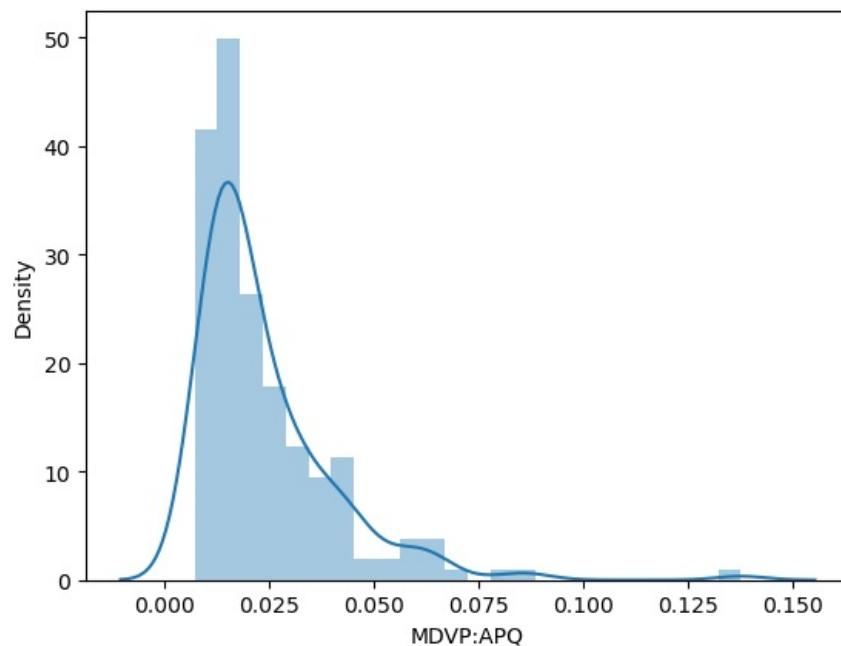
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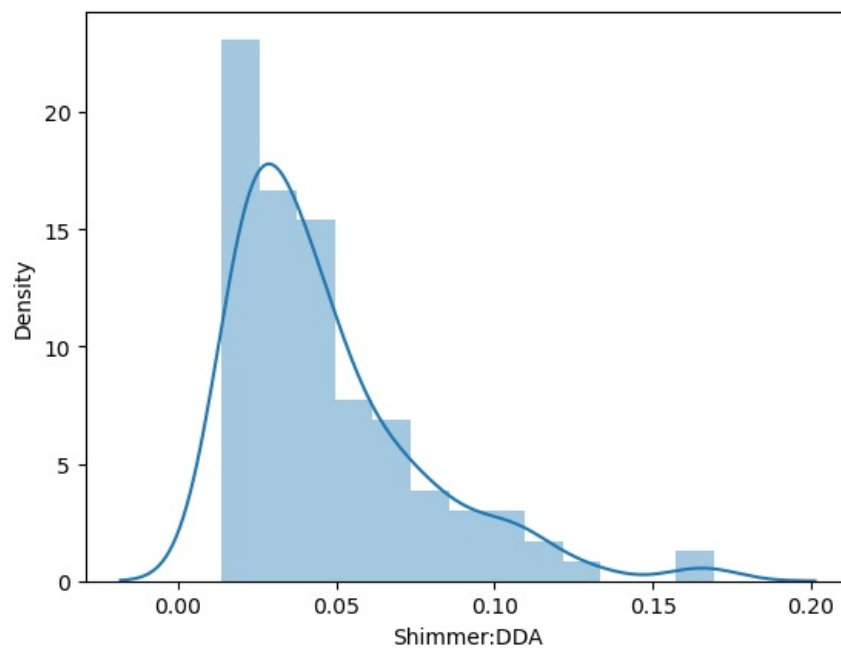
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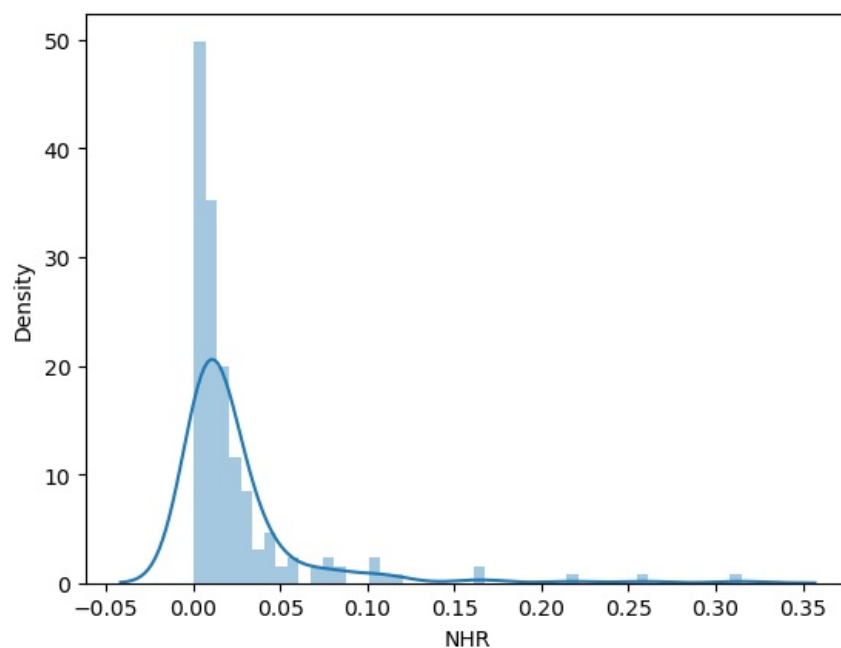
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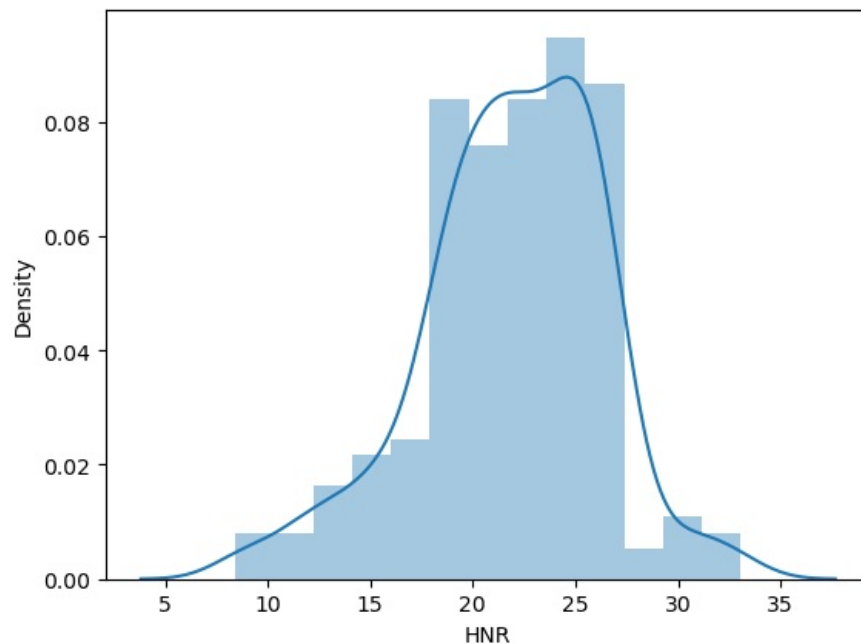
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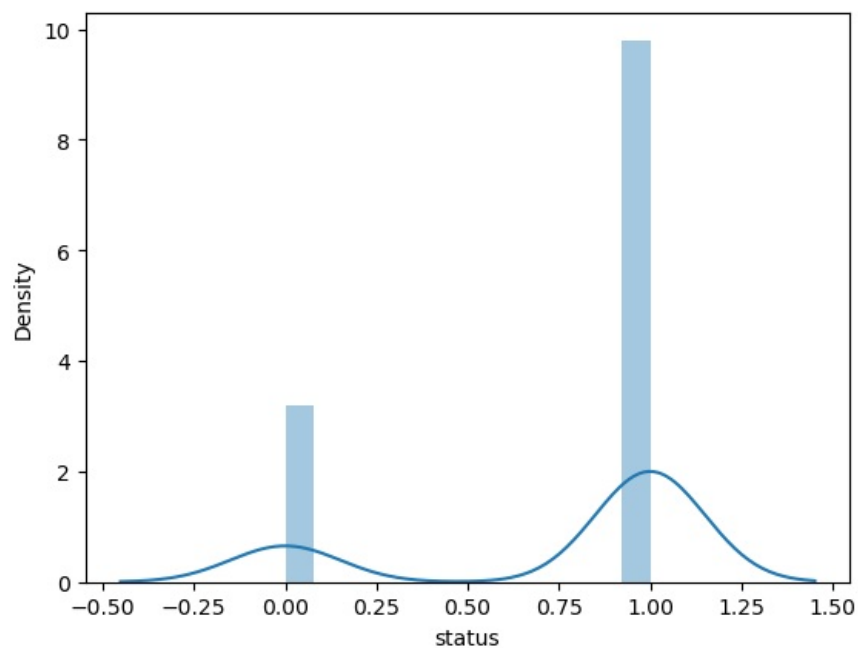
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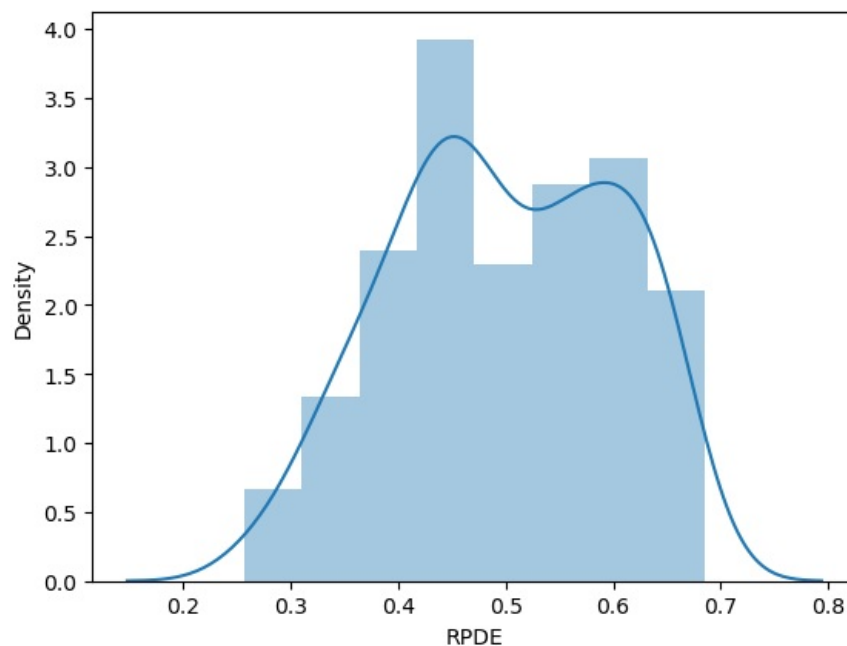
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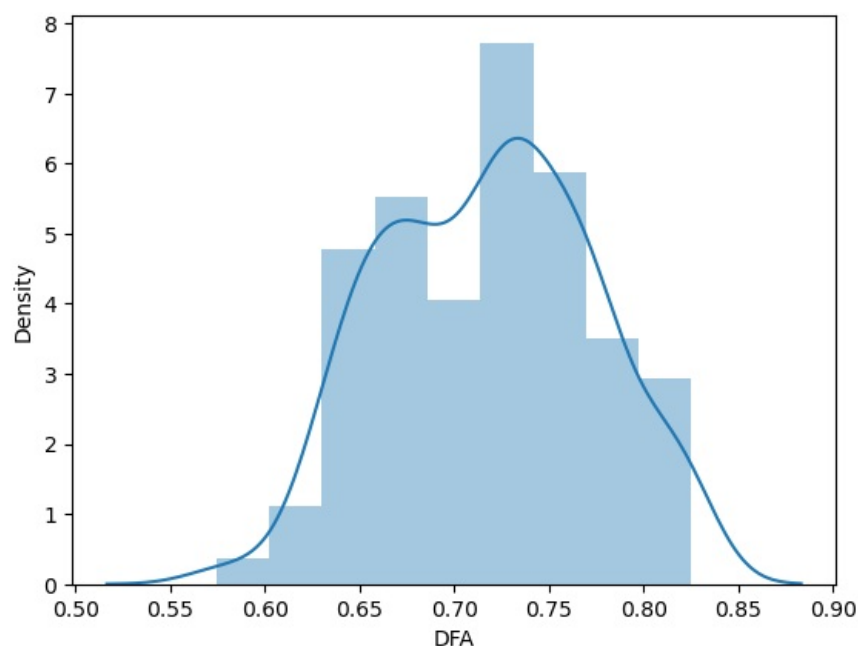
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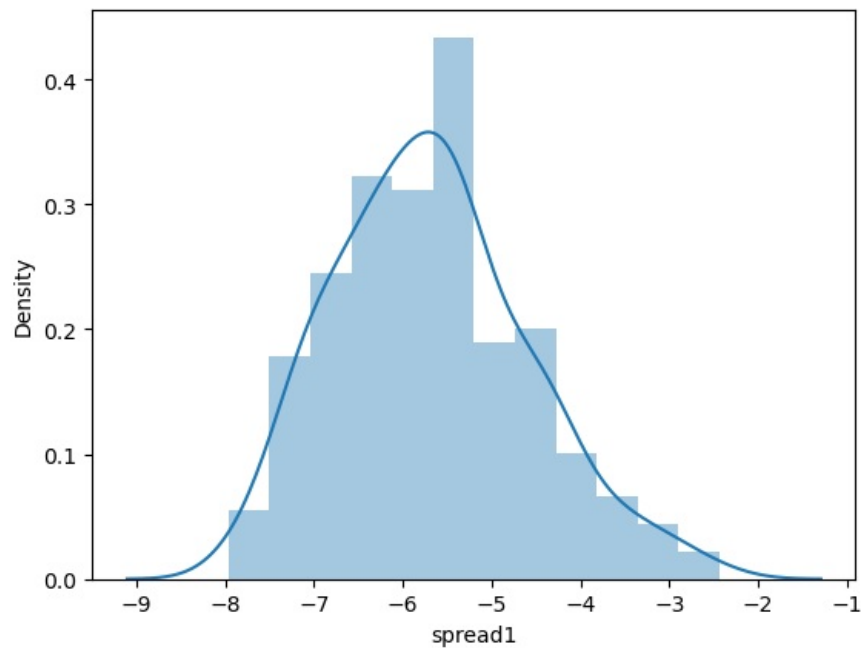
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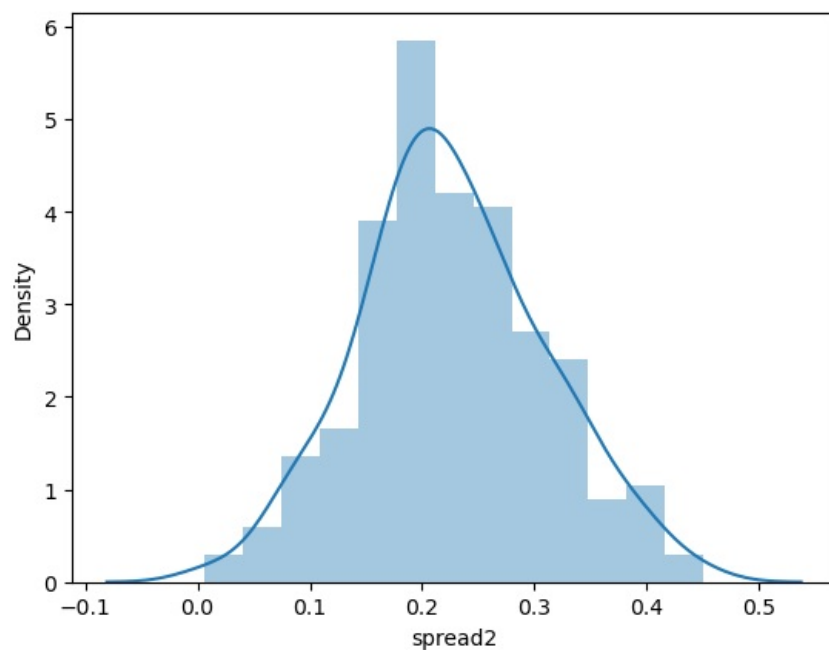
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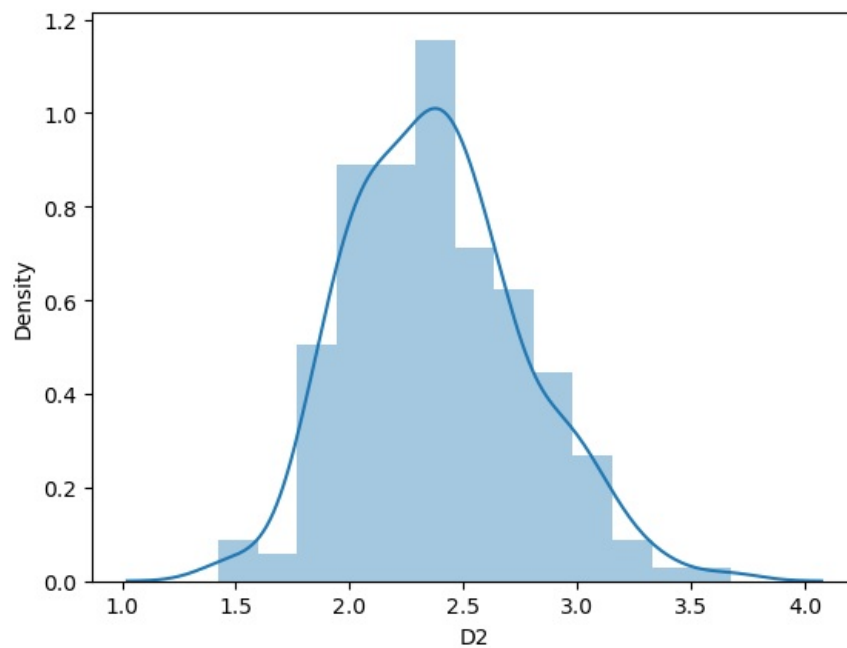
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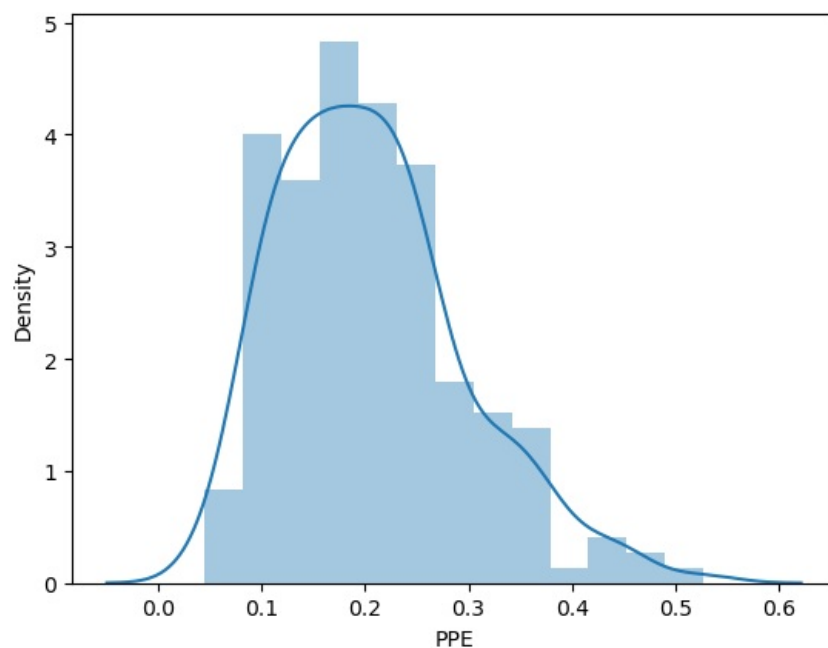
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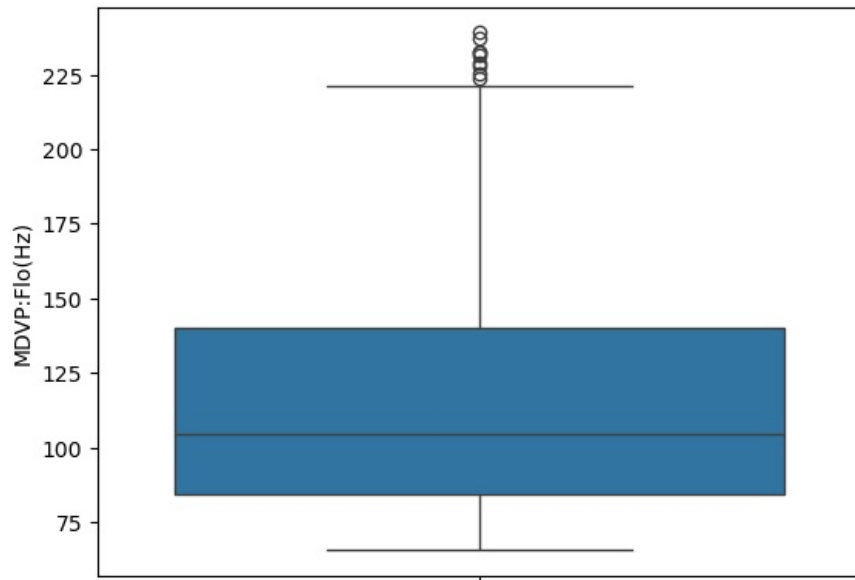
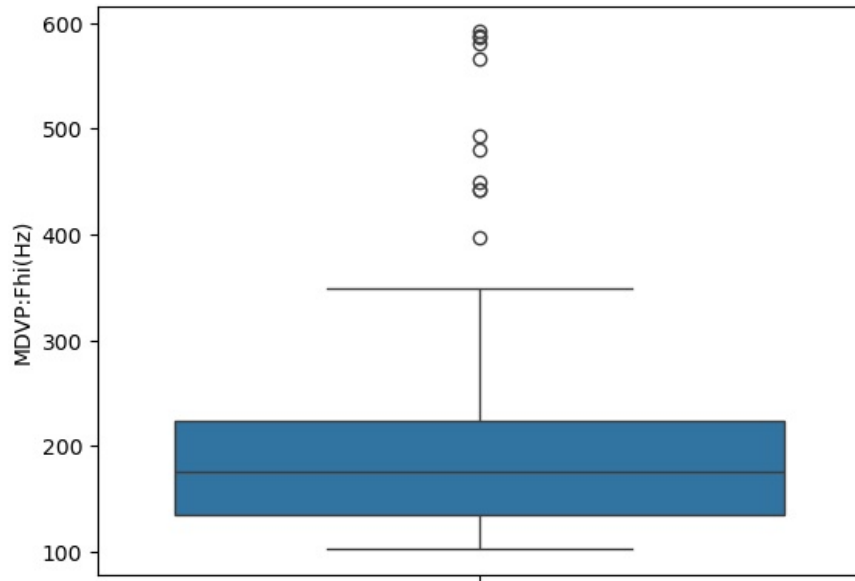
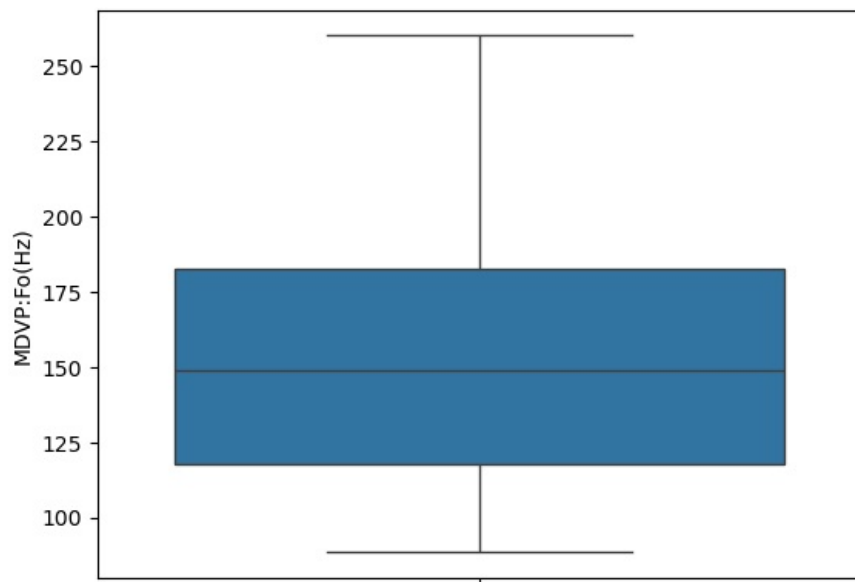
Please adapt your code to use either ``displot`` (a figure-level function with similar flexibility) or ``histplot`` (an axes-level function for histograms).

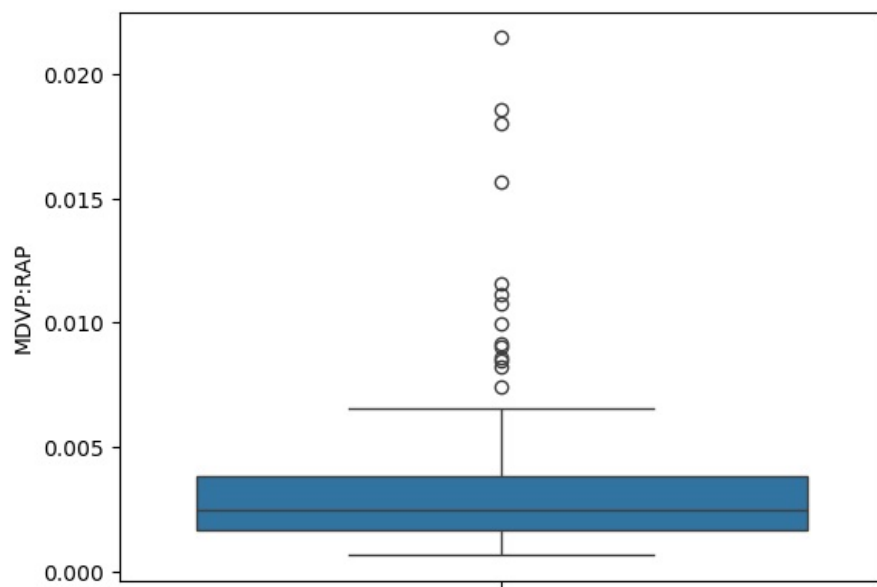
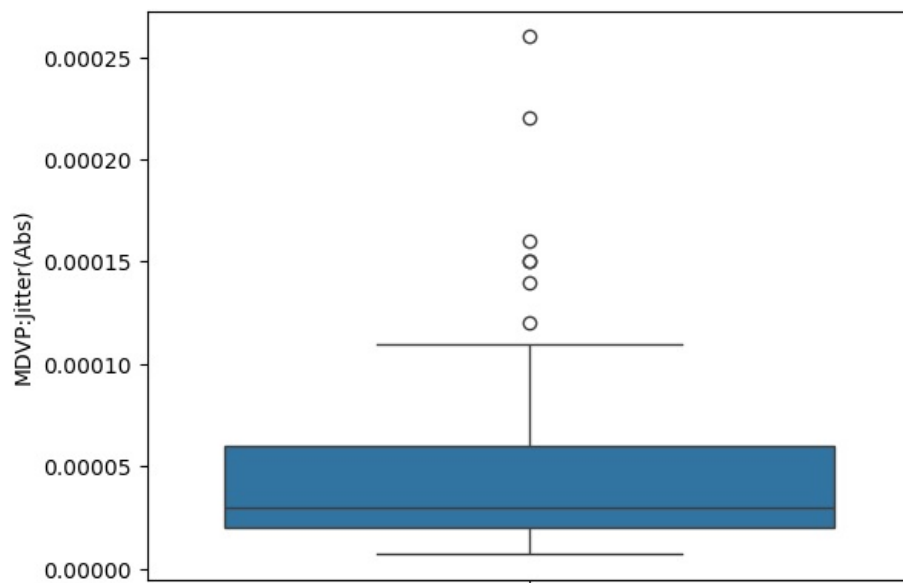
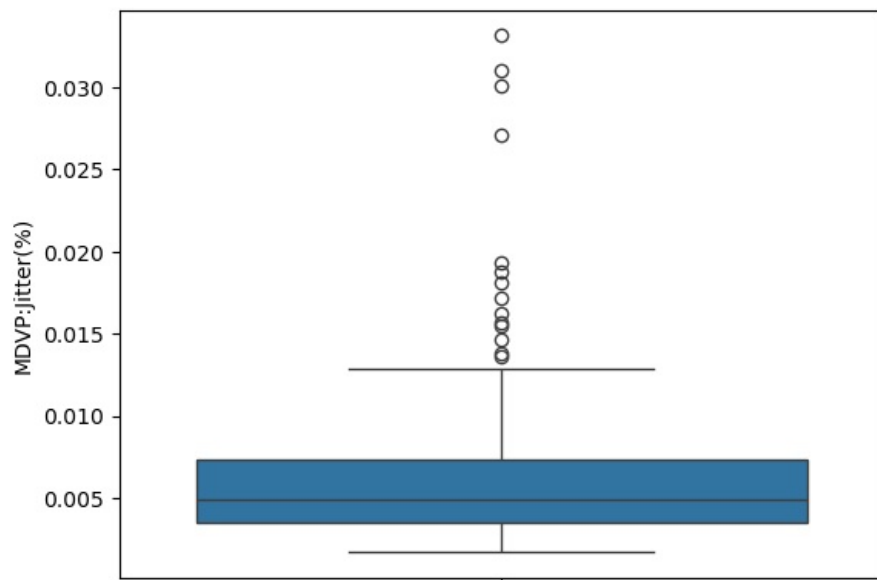
For a guide to updating your code to use the new functions, please see <https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751>

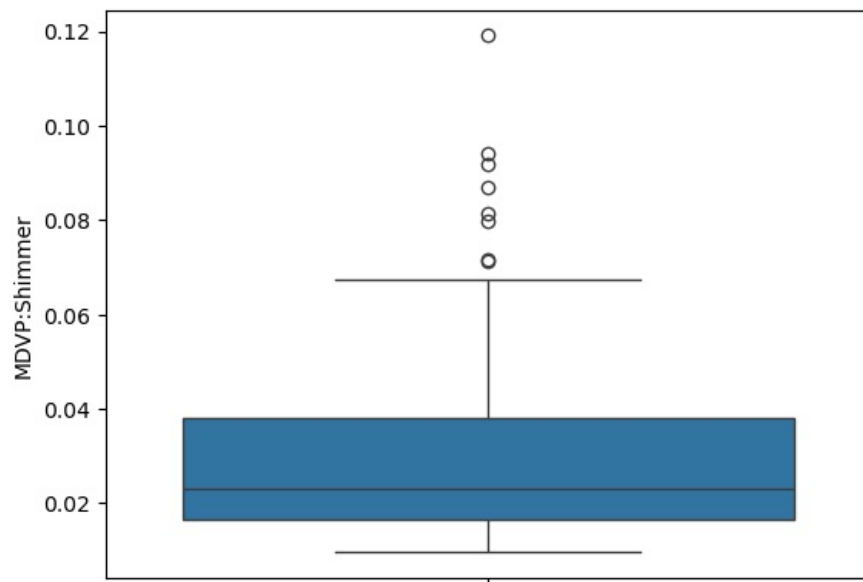
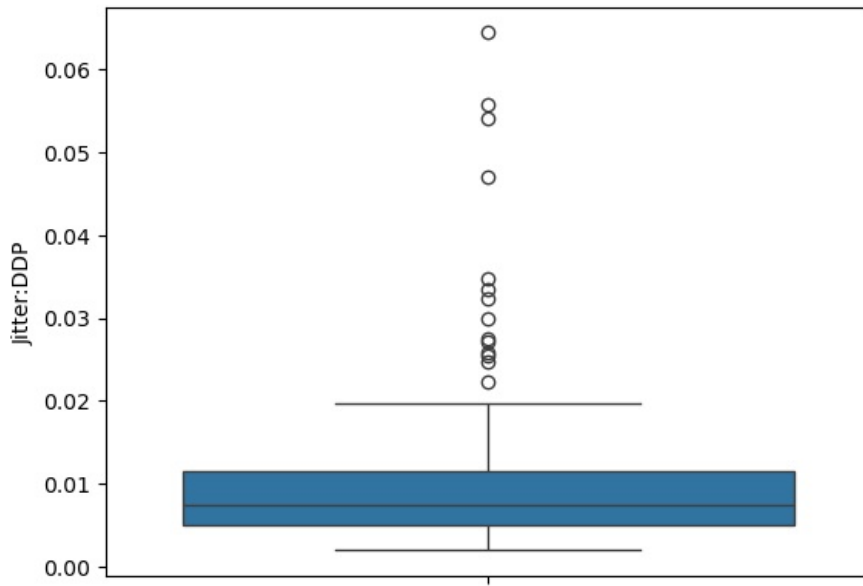
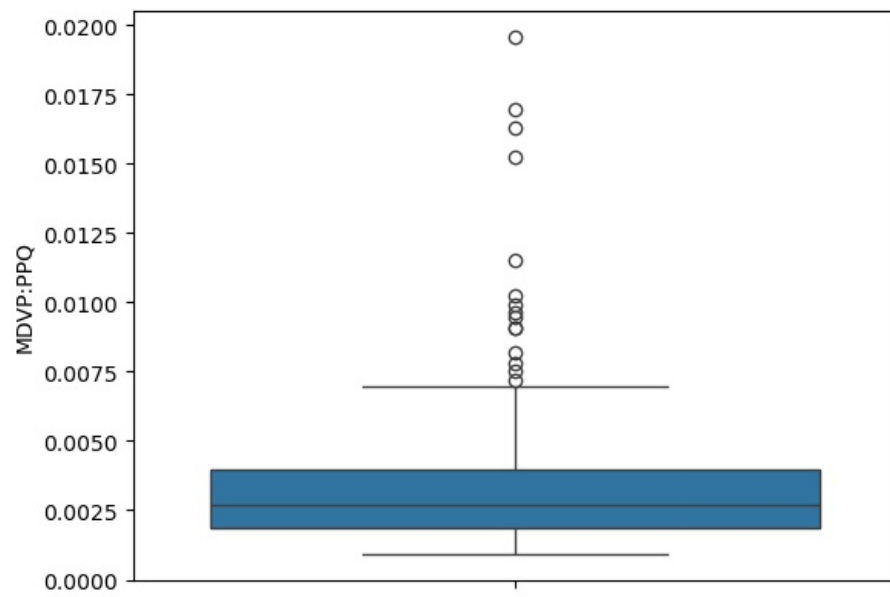
```
sns.distplot(df[col])
```

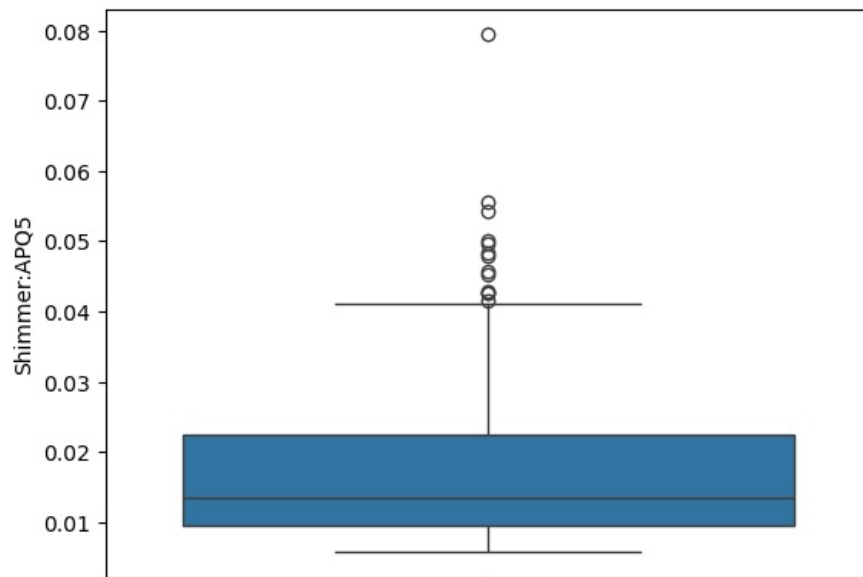
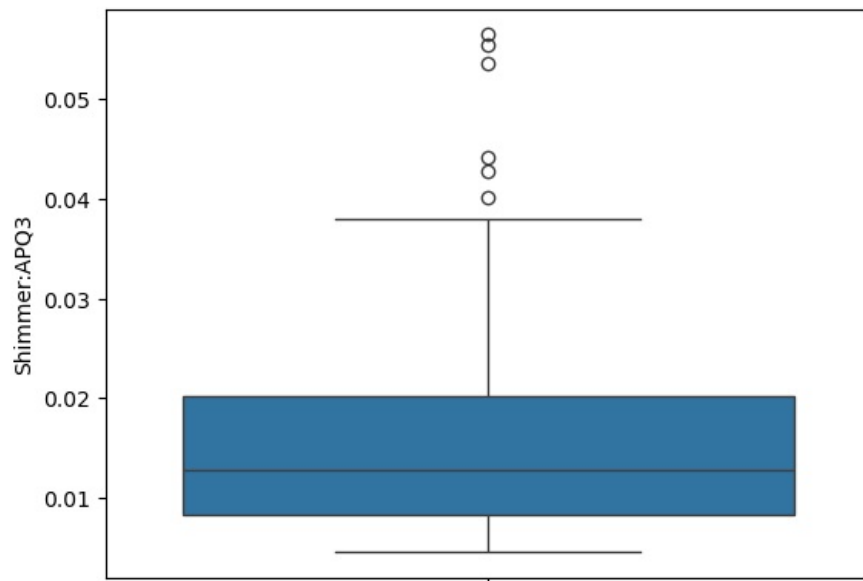
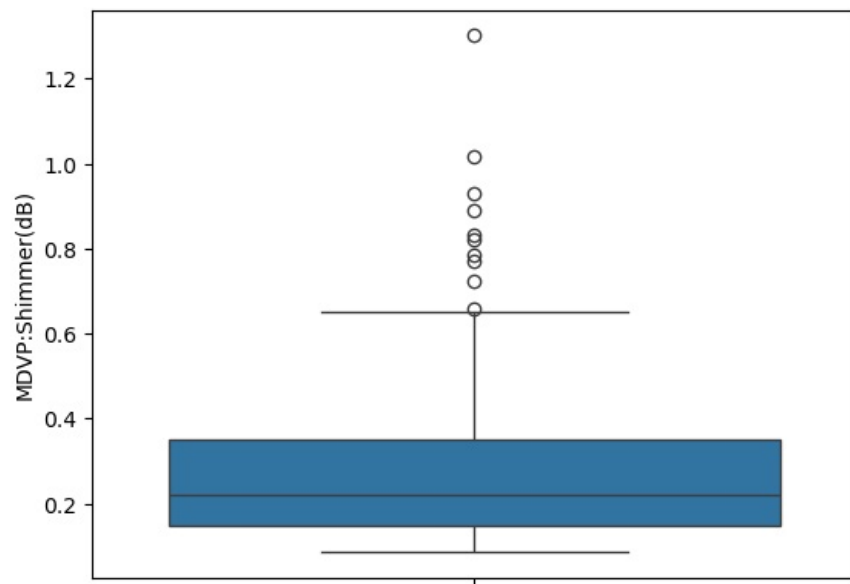


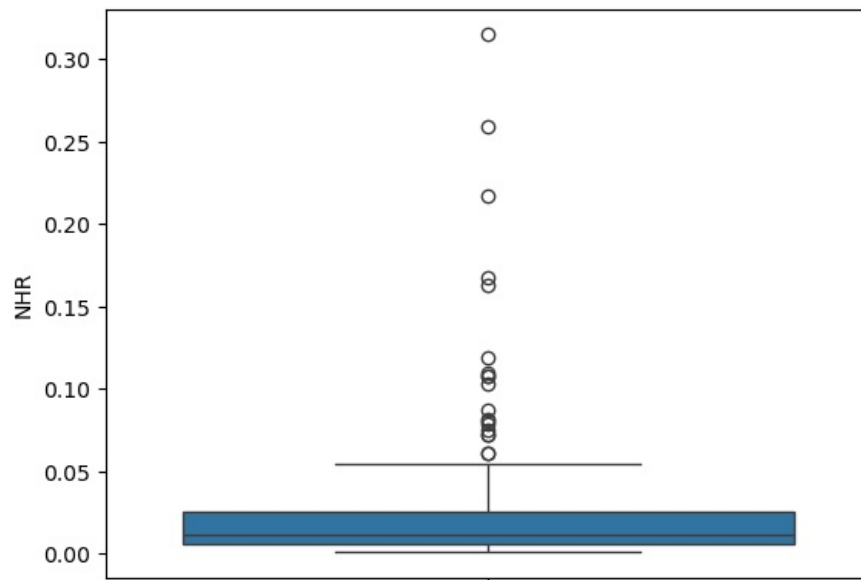
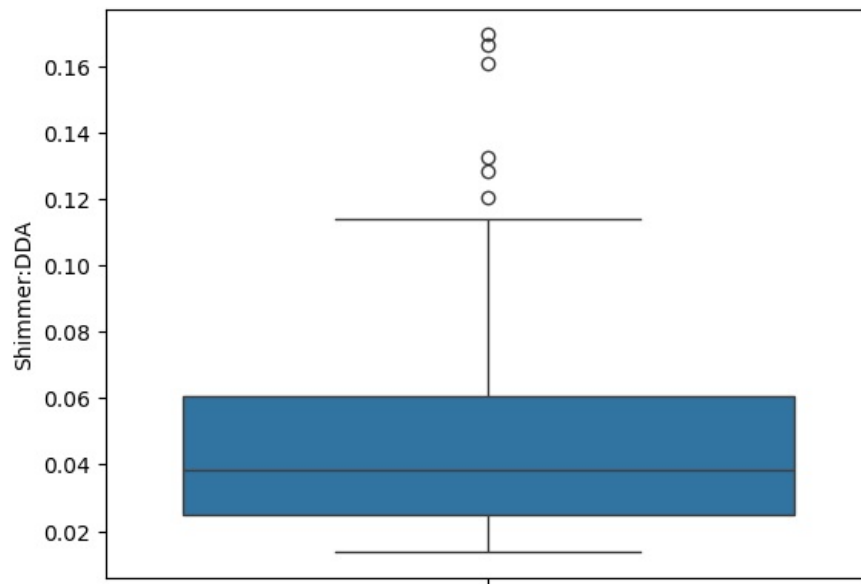
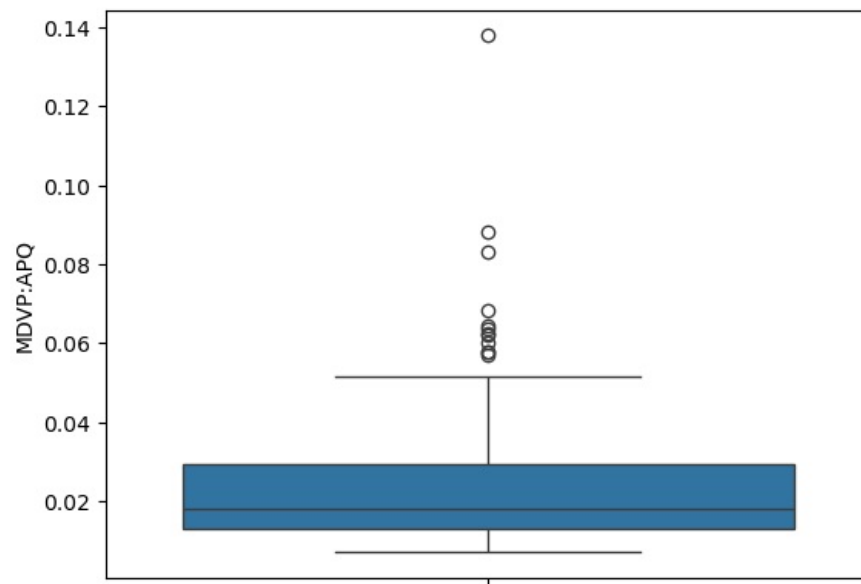
```
In [23]: def boxplots(col):  
          sns.boxplot(df[col])  
          plt.show()  
          for i in list(df.columns)[1:]:  
              boxplots(i)
```

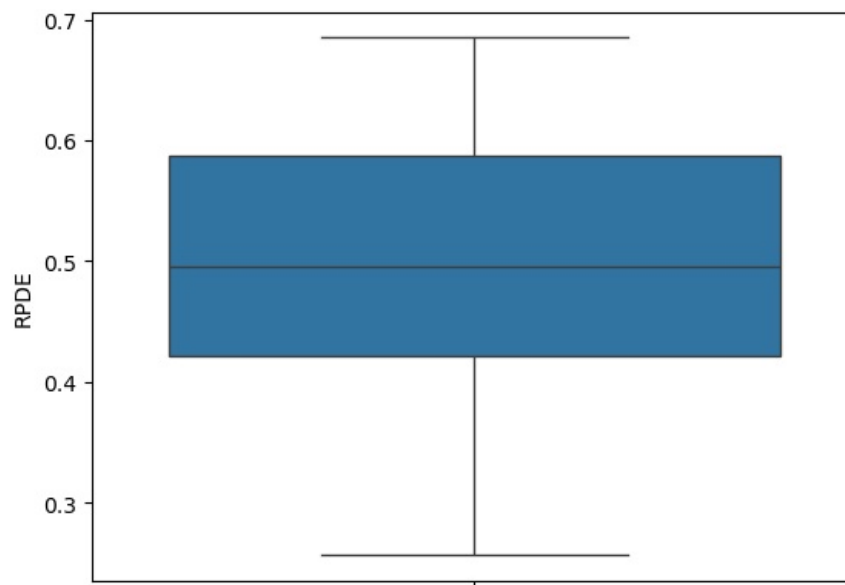
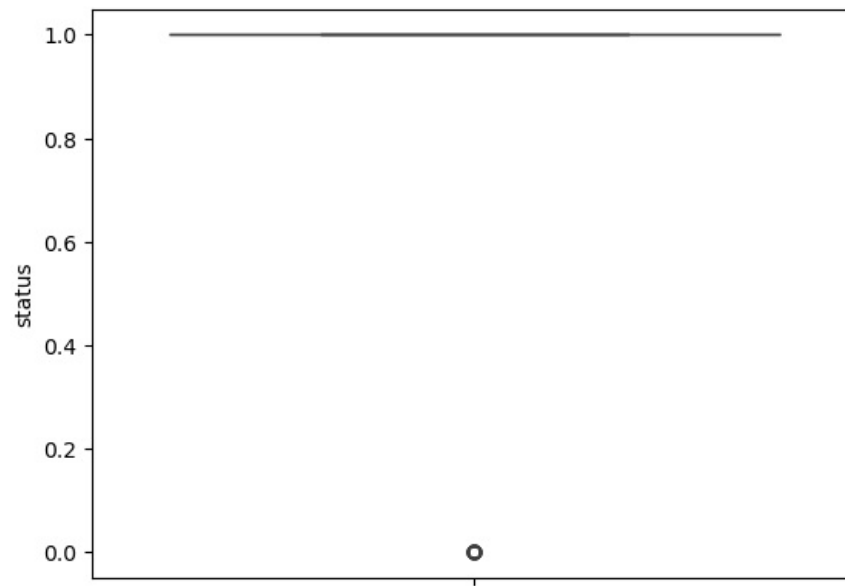
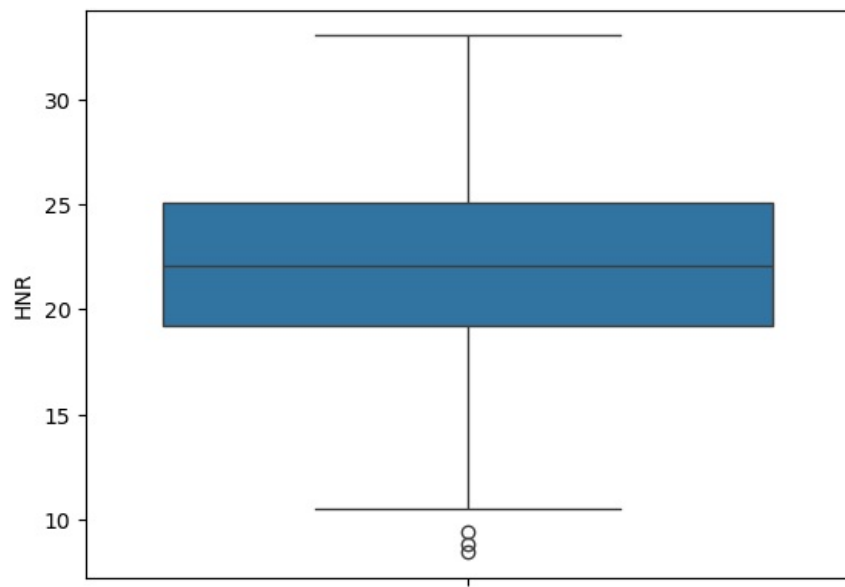


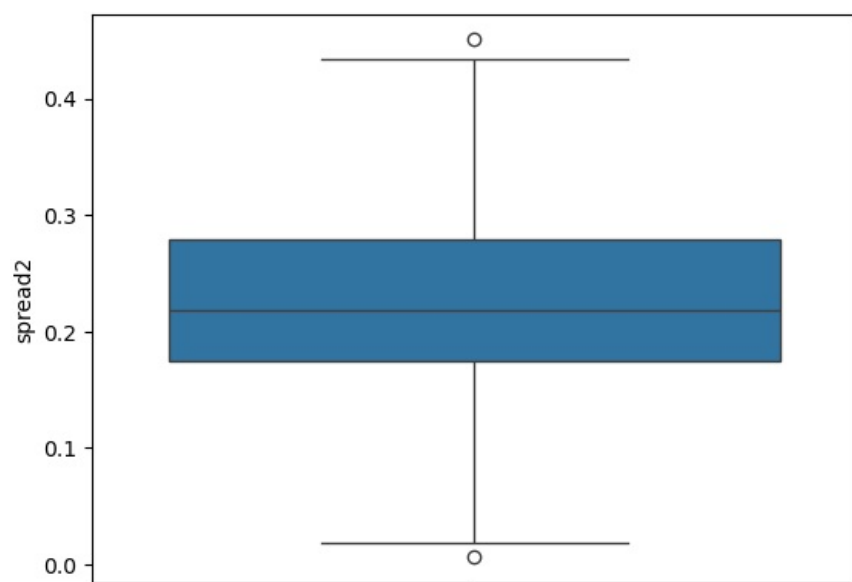
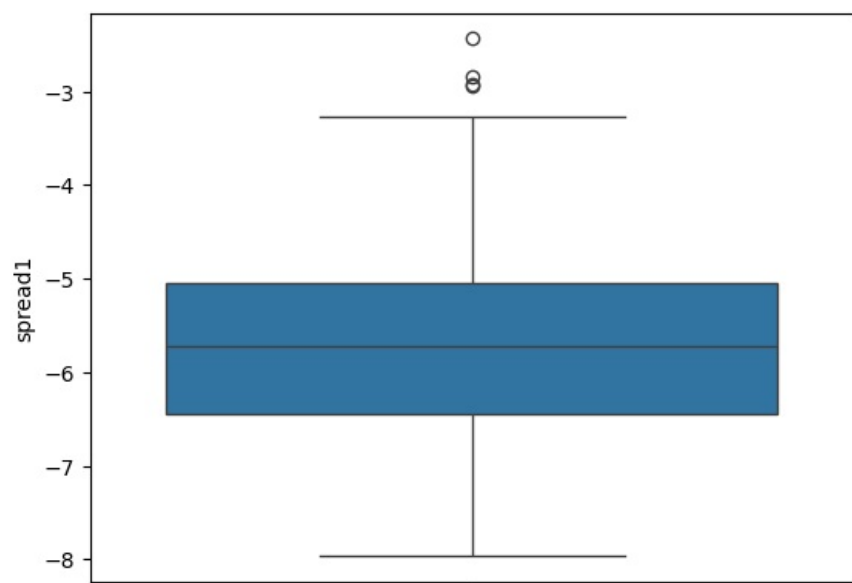
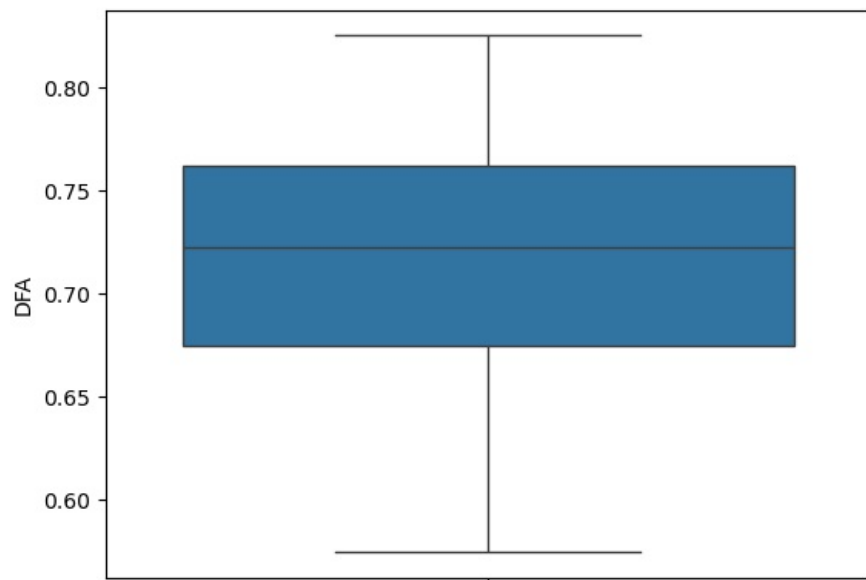


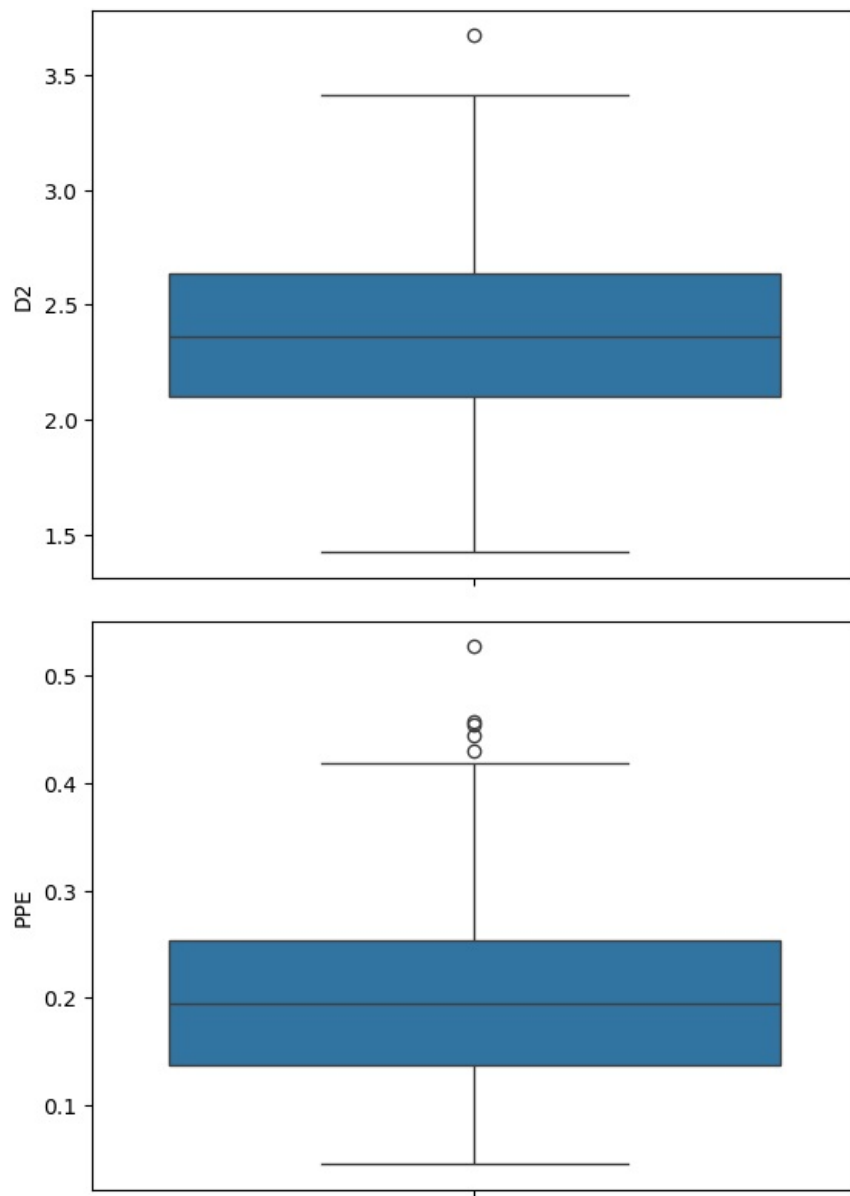






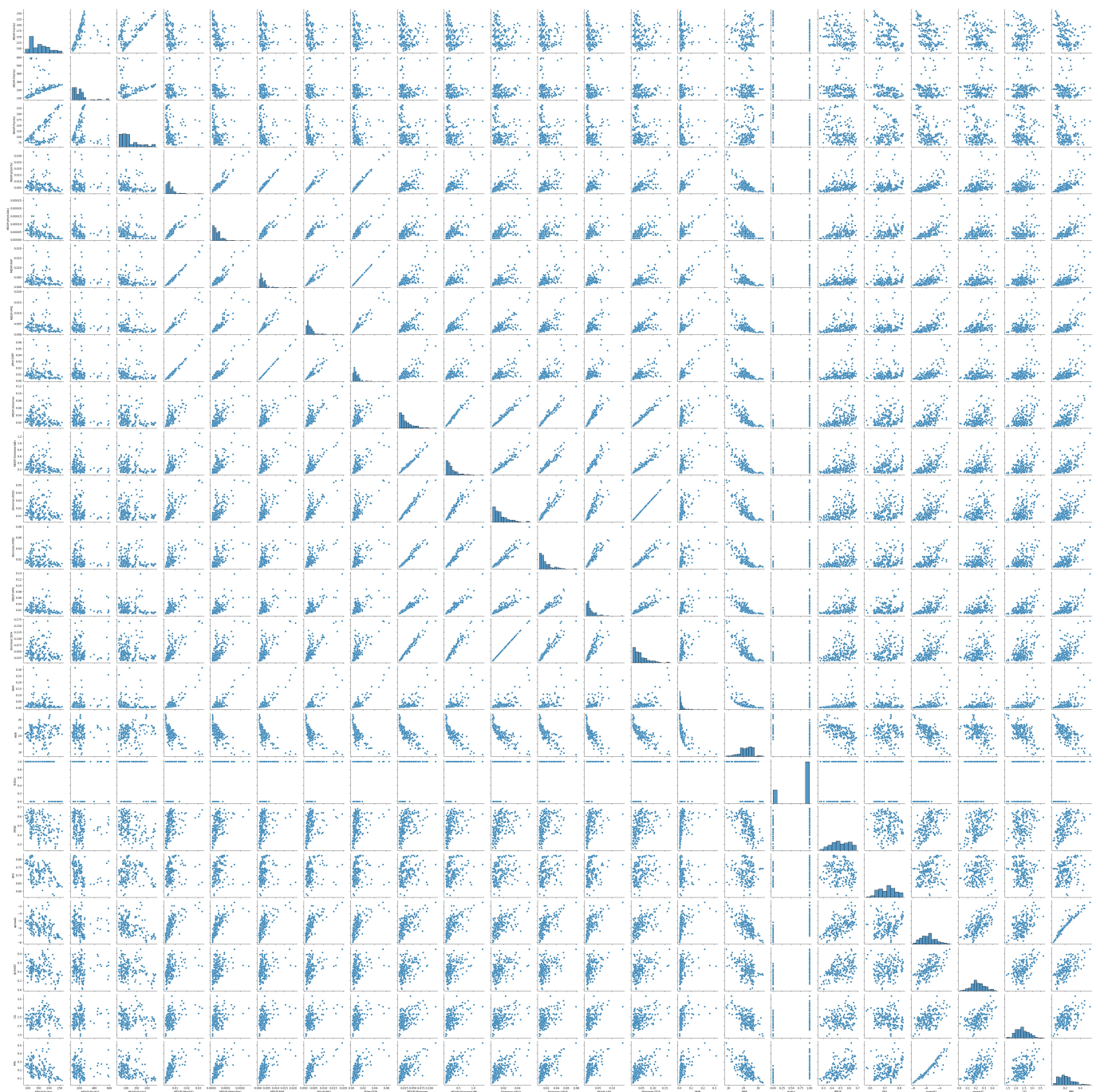






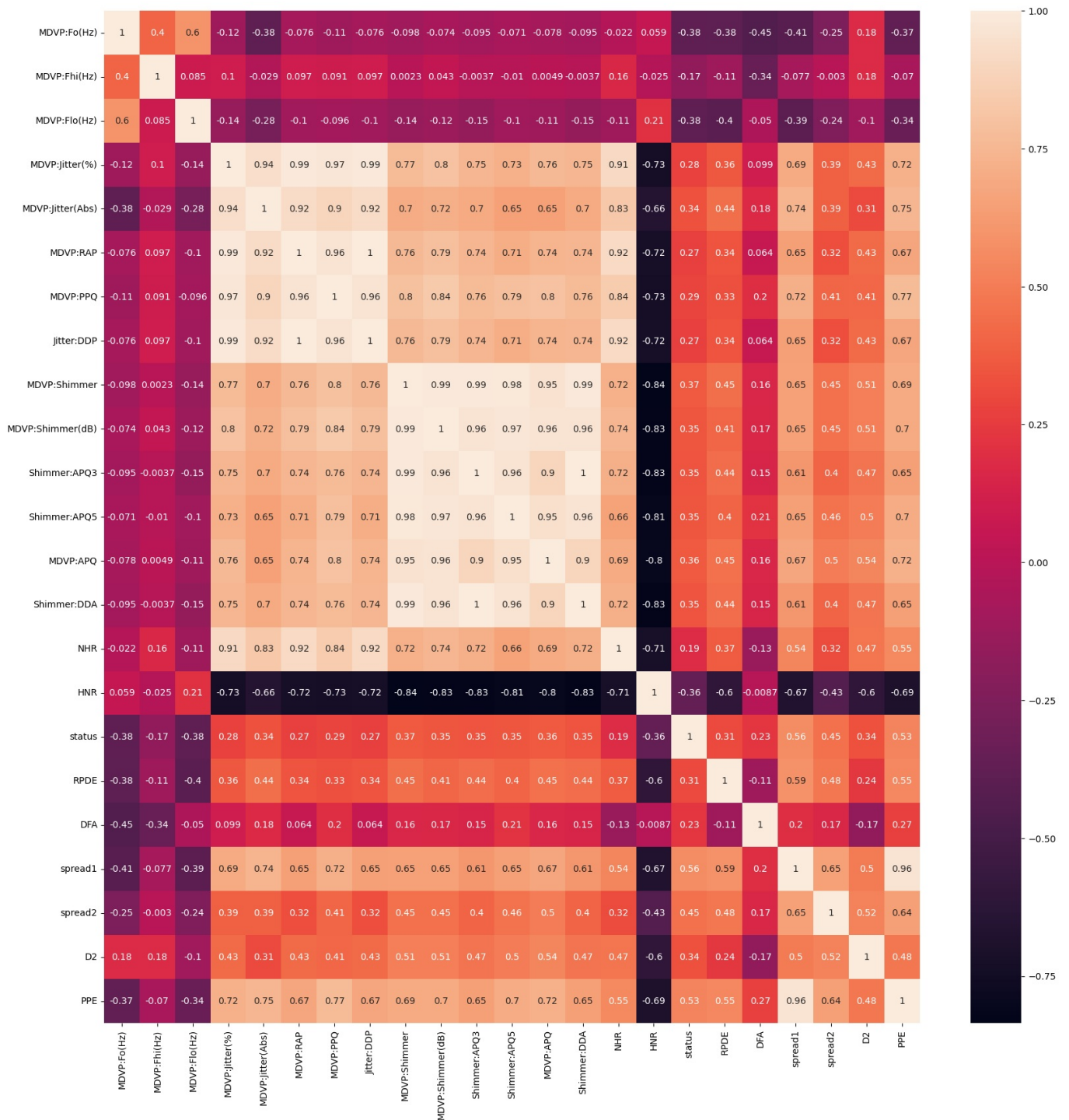
```
In [24]: sns.pairplot(df)
```

```
Out[24]: <seaborn.axisgrid.PairGrid at 0x1b5248b6900>
```



```
In [25]: # Find correlation
plt.figure(figsize=(20,20))
# Only select numeric columns for correlation
numeric_df = df.select_dtypes(include=['float64', 'int64'])
# Calculate correlation on numeric columns only
corr = numeric_df.corr()
# Plot the heatmap
sns.heatmap(corr, annot=True)
```

```
Out[25]: <Axes: >
```



```
In [21]: x=df.drop(columns=["name","status"],axis=1)
y=df["status"]
```

```
In [22]: x
```

Out[22]:

	MDVP:Fo(Hz)	MDVP:Fhi(Hz)	MDVP:Flo(Hz)	MDVP:Jitter(%)	MDVP:Jitter(Abs)	MDVP:RAP	MDVP:PPQ	Jitter:DDP	MDVP:Shim
0	119.992	157.302	74.997	0.00784	0.00007	0.00370	0.00554	0.01109	0.04
1	122.400	148.650	113.819	0.00968	0.00008	0.00465	0.00696	0.01394	0.06
2	116.682	131.111	111.555	0.01050	0.00009	0.00544	0.00781	0.01633	0.05
3	116.676	137.871	111.366	0.00997	0.00009	0.00502	0.00698	0.01505	0.05
4	116.014	141.781	110.655	0.01284	0.00011	0.00655	0.00908	0.01966	0.06
...	...	...	...	...	...	...	...	...	...
190	174.188	230.978	94.261	0.00459	0.00003	0.00263	0.00259	0.00790	0.04
191	209.516	253.017	89.488	0.00564	0.00003	0.00331	0.00292	0.00994	0.02
192	174.688	240.005	74.287	0.01360	0.00008	0.00624	0.00564	0.01873	0.02
193	198.764	396.961	74.904	0.00740	0.00004	0.00370	0.00390	0.01109	0.02
194	214.289	260.277	77.973	0.00567	0.00003	0.00295	0.00317	0.00885	0.01

195 rows × 22 columns



In [23]:

y

Out[23]:

```
0      1
1      1
2      1
3      1
4      1
..
190    0
191    0
192    0
193    0
194    0
Name: status, Length: 195, dtype: int64
```

In [24]:

```
from imblearn.over_sampling import RandomOverSampler
ros=RandomOverSampler()
x_ros,y_ros=ros.fit_resample(x,y)
print(y.value_counts())
print("*****")
print(y_ros.value_counts())
```

```
status
1      147
0       48
Name: count, dtype: int64
*****
status
1      147
0      147
Name: count, dtype: int64
```

Feature Scaling

In [25]:

```
from sklearn.preprocessing import MinMaxScaler

scaler=MinMaxScaler((-1,1))
x=scaler.fit_transform(x_ros)
y=y_ros
```

In [26]:

```
x.shape
```

Out[26]:

```
(294, 22)
```

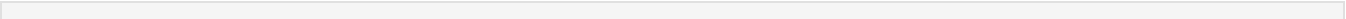
Applying PCA

In [27]:

```
from sklearn.decomposition import PCA

pca=PCA(0.95)
x_pca=pca.fit_transform(x)
print(x.shape)
print(x_pca.shape)
```

(294, 22)
(294, 8)



```
In [28]: from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(x_pca,y,test_size=0.2,random_state=2)
```

Model Building

```
In [29]: from sklearn.metrics import confusion_matrix,classification_report,accuracy_score
```

```
In [30]: list_met=[]
list_accuracy=[]
# Model 1 : Logistic Regression

from sklearn.linear_model import LogisticRegression
classifier=LogisticRegression(C=0.4,max_iter=1000,solver="liblinear")
lr=classifier.fit(x_train,y_train)
y_pred=classifier.predict(x_test)
# Accuracy
accuracy_LR=accuracy_score(y_test,y_pred)

# Model 2 : Decision Tree

from sklearn.tree import DecisionTreeClassifier
classifier2=DecisionTreeClassifier(random_state=14)
dt=classifier2.fit(x_train,y_train)
# Prediction
y_pred2=classifier2.predict(x_test)
# Evalution
accuracy_DT=accuracy_score(y_test,y_pred2)

# Model 3 : Random Forest Gini

from sklearn.ensemble import RandomForestClassifier
classifier3=RandomForestClassifier(random_state=14)
rf=classifier3.fit(x_train,y_train)
#Prediction
y_pred3=classifier3.predict(x_test)
#Evolution
accuracy_RF=accuracy_score(y_test,y_pred3)

# Model 4 : Random Forest Entropy

classifier4=RandomForestClassifier(criterion="entropy",random_state=14)
rfe=classifier4.fit(x_train,y_train)
#Prediction
y_pred4=classifier4.predict(x_test)
#Evolution
accuracy_RFE=accuracy_score(y_test,y_pred4)

#Model 5 : Support Vector Machine

from sklearn.svm import SVC
model_svm=SVC()
SVM=model_svm.fit(x_train,y_train)
#Prediction
y_pred5=model_svm.predict(x_test)
#Evolution
accuracy_SVC=accuracy_score(y_test,y_pred5)

# Model 6 : KNN

from sklearn.neighbors import KNeighborsClassifier
model_knn=KNeighborsClassifier()
knn=model_knn.fit(x_train,y_train)
#Prediction
y_pred6=model_knn.predict(x_test)
#Evolution
accuracy_KNN=accuracy_score(y_test,y_pred6)

# Model 7 : Gaussian Naive Bayes

from sklearn.naive_bayes import GaussianNB
gnb=GaussianNB()
gnb=gnb.fit(x_train,y_train)
#Prediction
y_pred7=gnb.predict(x_test)
#Evolution
accuracy_GNB=accuracy_score(y_test,y_pred7)

# Model 8 : Bernoulli Neive Bayes

from sklearn.naive_bayes import BernoulliNB
bnb=BernoulliNB()
```

```

bnb=bnb.fit(x_train,y_train)
#Prediction
y_pred7=bnb.predict(x_test)
#Evolution
accuracy_BNB=accuracy_score(y_test,y_pred7)

# COMBINE ALL THE MODELS BY USING VOTING CLASSIFIER MODEL

from sklearn.ensemble import VotingClassifier
evc=VotingClassifier(estimators=[("lr",lr),("DT",dt),("RF",rf),("RFE",rfe),("SVC",SVM),("KNN",knn),("GNB",gnb),

model_evc=evc.fit(x_train,y_train)
#Prediction
pred_evc=evc.predict(x_test)
#Evolution
accuracy_evc=accuracy_score(y_test,pred_evc)

list1=["Logistic Regreesion","Decison Tree","RandomForest Gini","RandomForest Entropy","Support Vector Machine"]
list2=[accuracy_LR,accuracy_DT,accuracy_RF,accuracy_RFE,accuracy_SVC,accuracy_KNN,accuracy_GNB,accuracy_BNB,accuracy_evc]
list3=[classifier,classifier2,classifier3,classifier4,model_svm,model_knn,gnb,bnb,model_evc]

df_accuracy=pd.DataFrame({"Method Used":list1,"Accuracy":list2})
print(df_accuracy)

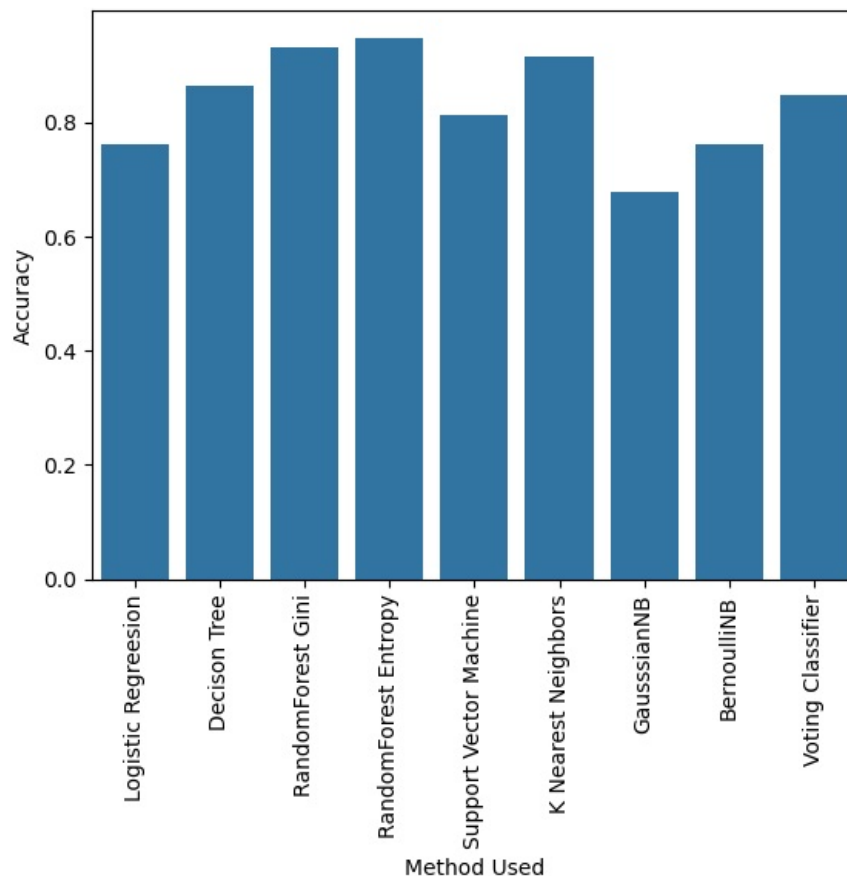
charts=sns.barplot(x="Method Used",y="Accuracy",data=df_accuracy)
charts.set_xticklabels(charts.get_xticklabels(),rotation=90)
print(charts)

```

	Method Used	Accuracy
0	Logistic Regreesion	0.762712
1	Decison Tree	0.864407
2	RandomForest Gini	0.932203
3	RandomForest Entropy	0.949153
4	Support Vector Machine	0.813559
5	K Nearest Neighbors	0.915254
6	GausssianNB	0.677966
7	BernoulliNB	0.762712
8	Voting Classifier	0.847458

Axes(0.125,0.11;0.775x0.77)

C:\Users\abhis\AppData\Local\Temp\ipykernel_24192\430780339.py:102: UserWarning: set_ticklabels() should only be used with a fixed number of ticks, i.e. after set_ticks() or using a FixedLocator.
charts.set_xticklabels(charts.get_xticklabels(),rotation=90)



In [31]: !pip install xgboost

Requirement already satisfied: xgboost in c:\users\abhis\anaconda3\lib\site-packages (3.0.4)
 Requirement already satisfied: numpy in c:\users\abhis\anaconda3\lib\site-packages (from xgboost) (2.1.3)
 Requirement already satisfied: scipy in c:\users\abhis\anaconda3\lib\site-packages (from xgboost) (1.15.3)

```
In [32]: from xgboost import XGBClassifier
model_xgb=XGBClassifier()
model_xgb.fit(x_train,y_train)
```

```
Out[32]: XGBClassifier
XGBClassifier(base_score=None, booster=None, callbacks=None,
              colsample_bylevel=None, colsample_bynode=None,
              colsample_bytree=None, device=None, early_stopping_rounds=None,
              enable_categorical=False, eval_metric=None, feature_types=None,
              feature_weights=None, gamma=None, grow_policy=None,
              importance_type=None, interaction_constraints=None,
              learning_rate=None, max_bin=None, max_cat_threshold=None,
              max_cat_to_onehot=None, max_delta_step=None, max_depth=None,
              max_leaves=None, min_child_weight=None, missing=nan,
```

```
In [33]: y_pred_xgb=model_xgb.predict(x_test)
print(accuracy_score(y_test,y_pred_xgb))
```

0.9152542372881356

```
In [34]: # OTHER EVOLUTION METHOD
from sklearn.metrics import roc_curve,auc
```

```
In [35]: # RandomForest Entropy
y_pred4_train=classifier4.predict(x_train)
y_pred4_test=classifier4.predict(x_test)

# KNN
pred_knn_train=model_knn.predict(x_train)
pred_knn_test=model_knn.predict(x_test)
```

```
In [36]: print(confusion_matrix(y_train,y_pred4_train))
print("*****5")
print(confusion_matrix(y_test,y_pred4_test))
```

```
[[120  0]
 [ 0 115]]
*****
[[27  0]
 [ 3 29]]
```

```
In [37]: print(classification_report(y_train,y_pred4_train))
print("*****5")
print(classification_report(y_test,y_pred4_test))
```

	precision	recall	f1-score	support
0	1.00	1.00	1.00	120
1	1.00	1.00	1.00	115
accuracy			1.00	235
macro avg	1.00	1.00	1.00	235
weighted avg	1.00	1.00	1.00	235

	precision	recall	f1-score	support
0	0.90	1.00	0.95	27
1	1.00	0.91	0.95	32
accuracy			0.95	59
macro avg	0.95	0.95	0.95	59
weighted avg	0.95	0.95	0.95	59

```
In [38]: print(confusion_matrix(y_train,pred_knn_train))
print("*****5")
print(confusion_matrix(y_test,pred_knn_test))
```

```
[[120  0]
 [ 8 107]]
*****
[[27  0]
 [ 5 27]]
```

```
In [39]: print(classification_report(y_train,pred_knn_train))
```

```
print("*****5)")
print(classification_report(y_test,pred_knn_test))
```

	precision	recall	f1-score	support
0	0.94	1.00	0.97	120
1	1.00	0.93	0.96	115
accuracy			0.97	235
macro avg	0.97	0.97	0.97	235
weighted avg	0.97	0.97	0.97	235

```
*****
```

	precision	recall	f1-score	support
0	0.84	1.00	0.92	27
1	1.00	0.84	0.92	32
accuracy			0.92	59
macro avg	0.92	0.92	0.92	59
weighted avg	0.93	0.92	0.92	59

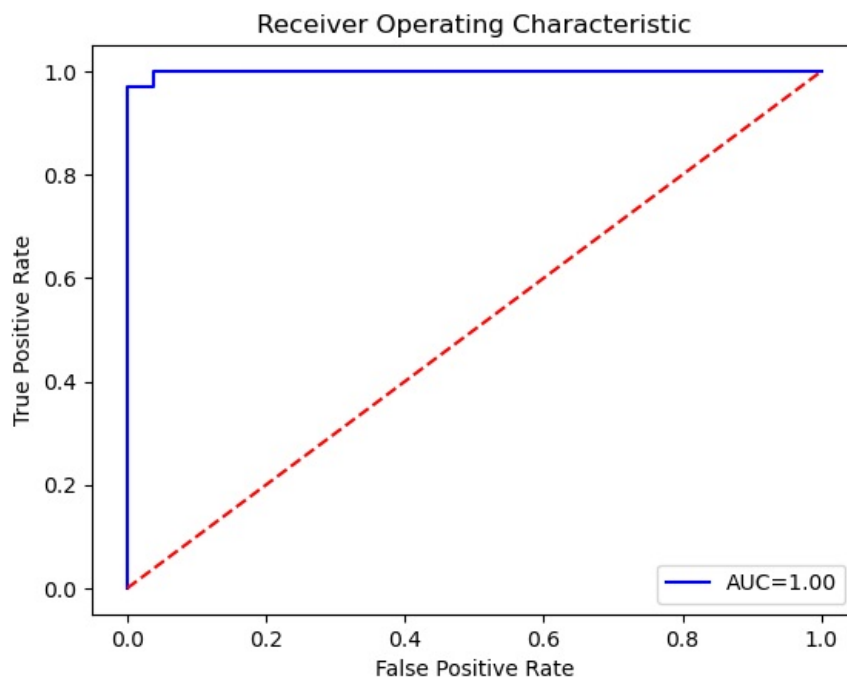
ROC AND AUC

```
In [41]: # Import the necessary library first
from sklearn.metrics import roc_curve, auc

# Define the plot_roc function if it's not already defined
def plot_roc(model, x_test, y_test):
    # Get predictions
    probabilities = model.predict_proba(np.array(x_test))
    predictions = probabilities
    fpr, tpr, threshold = roc_curve(y_test, predictions[:,1])
    roc_auc = auc(fpr, tpr)

    # Plot
    plt.figure()
    plt.plot(fpr, tpr, "b", label="AUC=%0.2f" % roc_auc)
    plt.plot([0, 1], [0, 1], 'r--')
    plt.xlabel('False Positive Rate')
    plt.ylabel('True Positive Rate')
    plt.title('Receiver Operating Characteristic')
    plt.legend(loc='lower right')
    plt.show()

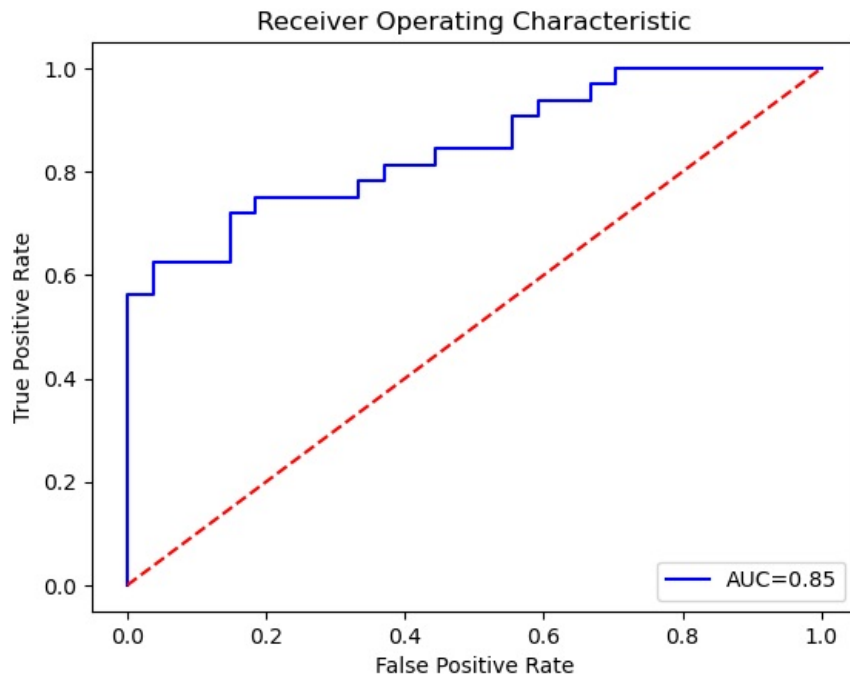
# Now call the function
plot_roc(rfe, x_test, y_test)
```



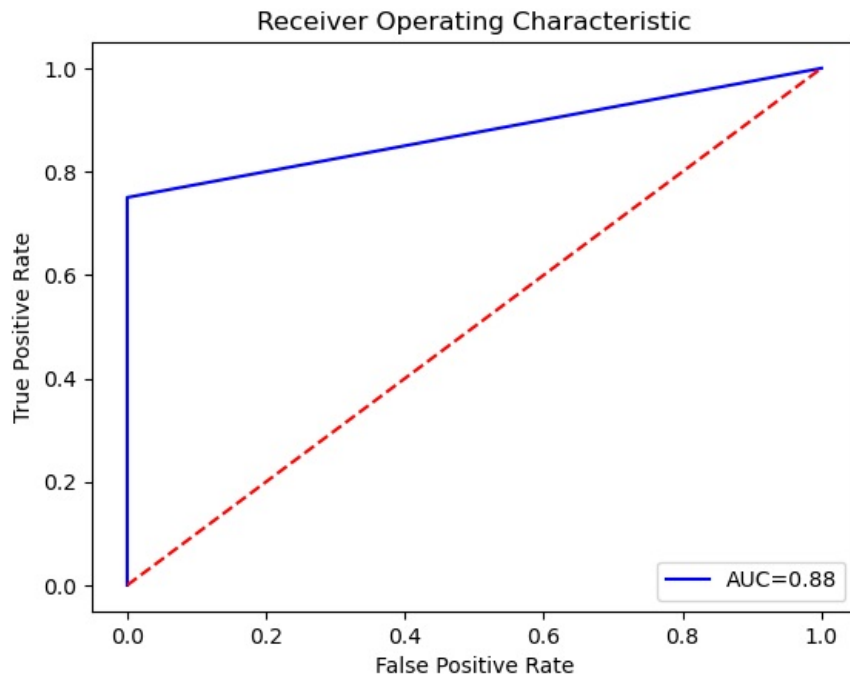
```
In [42]: for i in range(0, len(list3)):
    try:
        print()
        print("----- ROC FOR ", list1[i], " + PCA -----")
        plot_roc(list3[i], x_test, np.array(y_test))
        print()
```

```
except:
    print("ROC NOT VALID")
```

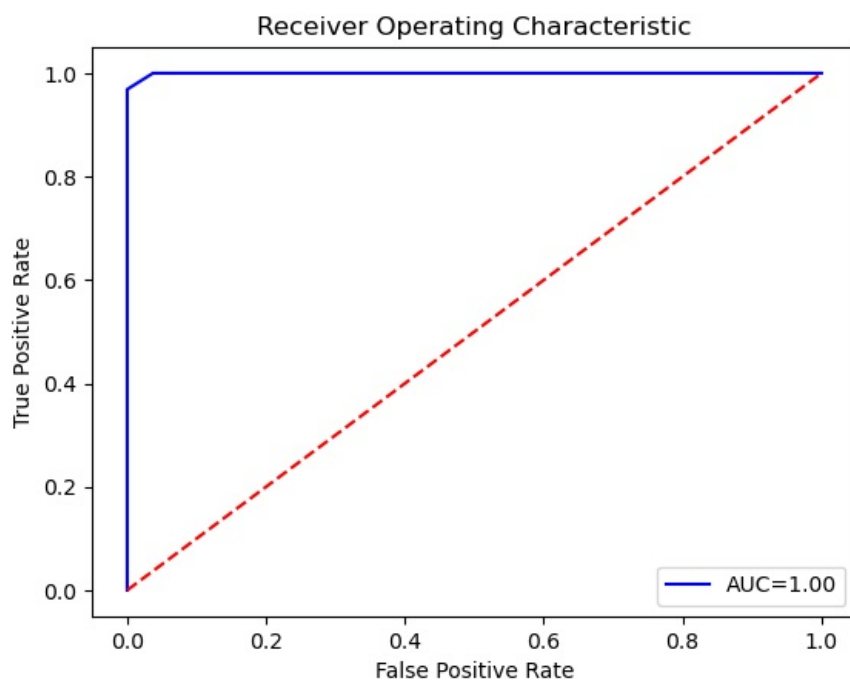
----- ROC FOR Logistic Regreesion + PCA -----



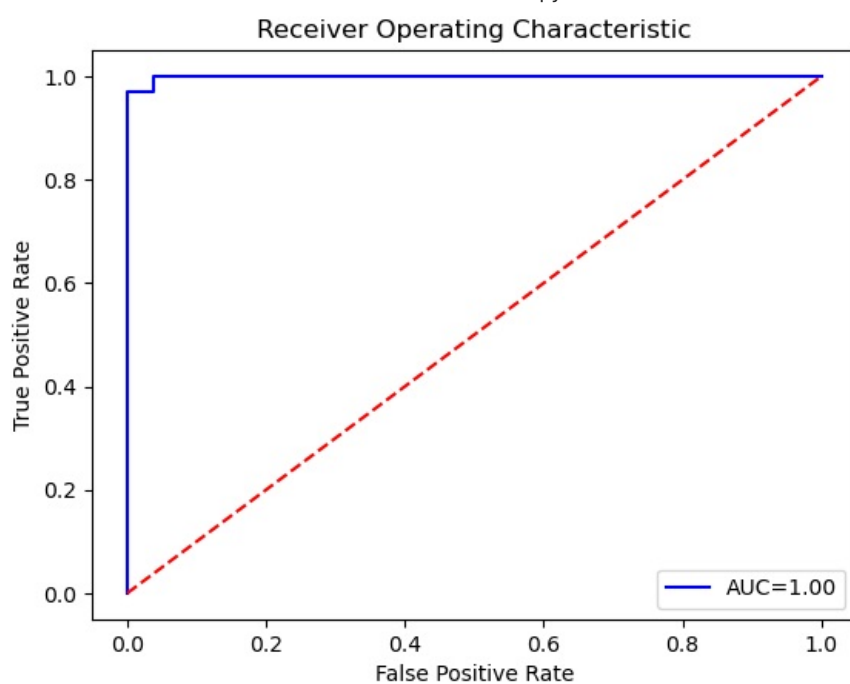
----- ROC FOR Decison Tree + PCA -----



----- ROC FOR RandomForest Gini + PCA -----

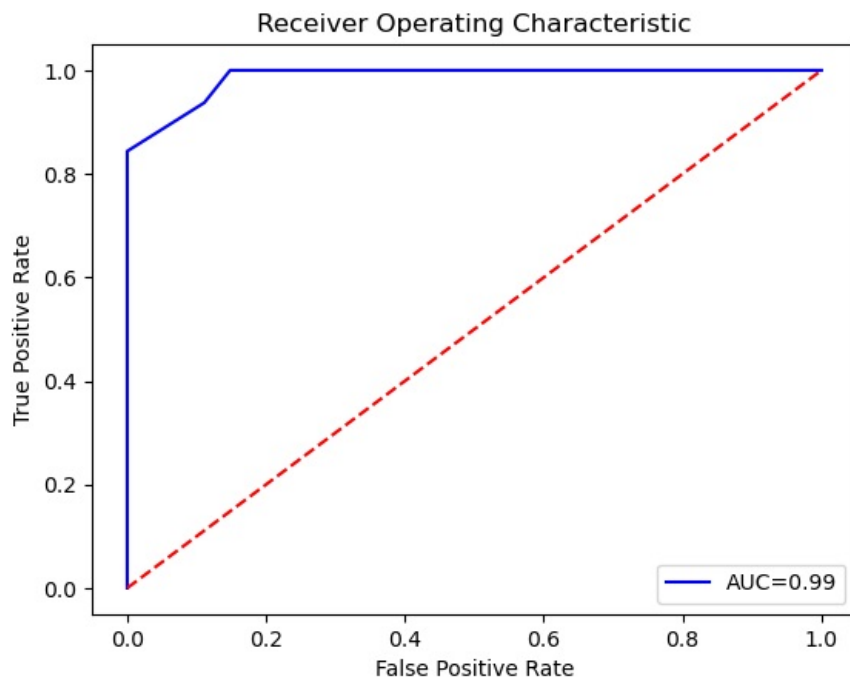


----- ROC FOR RandomForest Entropy + PCA -----

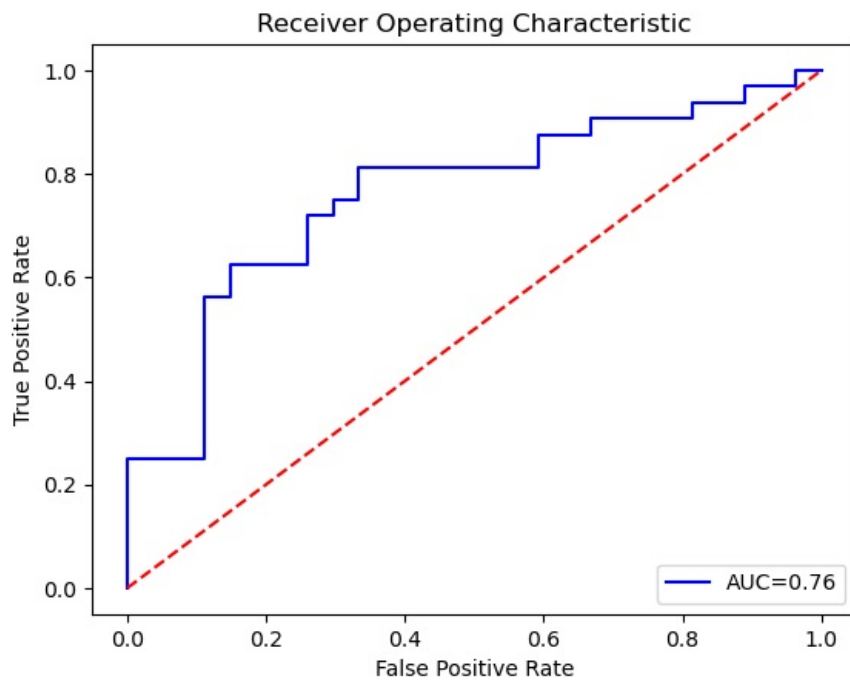


----- ROC FOR Support Vector Machine + PCA -----
ROC NOT VALID

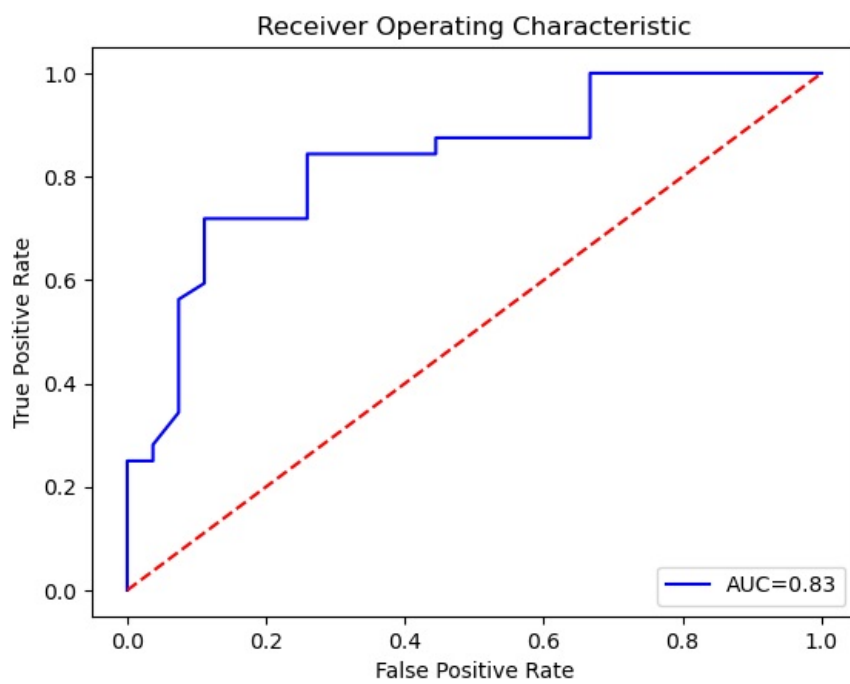
----- ROC FOR K Nearest Neighbors + PCA -----



----- ROC FOR GaussianNB + PCA -----



----- ROC FOR BernoulliNB + PCA -----



----- ROC FOR Voting Classifier + PCA -----
ROC NOT VALID

In []: